

Differentiable Simulation for Electrochemical Inverse Problems



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A dissertation submitted for the degree of
Masters in Mathematical Sciences

Trinity 2026

To my grandfather Ian.

Abstract

We develop a differentiable forward solver for voltammetric simulation and integrate it with the No-U-Turn (NUTS) sampler to perform Bayesian inference over electrochemical kinetic parameters. A custom adjoint rule for banded linear solves yields exact gradients from one adjoint system per timestep, caching only the matrix diagonals and solution vector rather than the full factorisation tape. This reduces the per-timestep memory requirements and eliminates the runtime overhead of tape traversal. The framework is applied to three reaction mechanisms of increasing complexity: a single electron transfer, a heterogeneous reaction, and an adsorption reaction. Under equal runtime budgets, NUTS produces more informative samples and converges to stable posterior estimates faster than random-walk Metropolis–Hastings. Both samplers recover the true parameters in all cases with the posterior widths and correlations encoding information that point estimates cannot.

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Chapter 1

Introduction

The inverse problem requires recovering physical parameters from indirect measurements. This problem is ubiquitous throughout the sciences; occurring whenever quantities of interest cannot be observed directly. In electrochemistry, the parameters governing a reaction must be inferred from the current response. This inference is the electrochemical inverse problem.

The corresponding forward problem is predicting the current response for a given set of parameters. In the present context this reduces to solving a diffusion PDE with kinetic boundary conditions. Solving the inverse problem then amounts to finding the parameters whose forward solution best explains the observed data. The difficulty is threefold: the mapping from parameters to current is nonlinear, distinct parameter configurations can produce similar current responses, and in certain kinetic regimes parameters may cease to influence the current altogether. Point estimates cannot capture these difficulties. They provide no account of uncertainty, nor can they reveal correlations between parameters. Bayesian inference addresses this by computing the posterior distribution conditioned on the observed data [11].

The standard tool for sampling such posteriors is Markov chain Monte Carlo. In this setting each likelihood evaluation requires a PDE solve, so every proposed sample, whether accepted or rejected, is expensive. Gradient-free samplers such as random-walk Metropolis–Hastings (RWMH) explore the parameter space by undirected local proposals and consequently waste many of these solves on rejected candidates, particularly as the dimensionality of parameter space grows. Gradient information helps in two ways: it can initialise the sampler in high density areas via optimisation; and it can inform the sampling itself with methods like Hamiltonian Monte Carlo (HMC), which uses the gradient to construct long-range, low-rejection proposals.

The challenge is obtaining these gradients cheaply. Finite differences require at least one additional forward solve per parameter and introduce truncation error.

Whereas, reverse-mode automatic differentiation computes exact gradients for all parameters at a cost bounded by a small multiple of the forward solve [16]. However, naïvely taping an entire PDE solver can be inefficient since it records every intermediate of the linear solve at each timestep. We instead derive a custom adjoint rule for the banded linear solve and register it with an automatic differentiation framework, *JAX* [2], so that gradient computation uses only the matrix diagonals and solution vector at each timestep.

This work sits within a broader shift toward differentiable scientific computing; Kitchin et al. [22] characterise differentiable programming as a fifth paradigm for modelling in chemical engineering. Concurrently, Chen et al. [5] have developed differentiable electrochemical simulators for gradient-based parameter recovery. Our work is complementary: we adopt a Bayesian framework rather than optimisation alone, and we compute gradients via a custom adjoint rather than taping the full solver. The Bayesian approach to voltammetric inference via MCMC was established by Gavaghan et al. [11]; we extend their framework by replacing the gradient-free sampler with gradient-informed alternatives enabled by the differentiable solver.

This dissertation makes three contributions. First, we develop a differentiable forward solver for voltammetric PDEs, in which a custom adjoint rule for the banded linear system provides exact gradients at low memory cost and generalises to the wider banded systems arising in multi-species reactions. Second, we integrate this solver with the No-U-Turn sampler (NUTS) [19], an extension of HMC, to obtain posterior distributions over the physical parameters. Third, we apply the framework to three reaction mechanisms of increasing complexity: a single electron transfer (3 parameters), a heterogeneous reaction (7 parameters), and an adsorption reaction (9 parameters). We find that the advantage of NUTS over RWMH grows with the dimensionality of the parameter space.

We develop the methodology in Chapters 2 to 6 using the single-electron-transfer reaction as running example to introduce each concept within the electrochemical context. Chapters 7 and 8 then apply the methodology to chemically richer and more complex reactions.

Chapter 2

Electrochemical Systems

In a typical electrochemical experiment, a *potentiostat* applies a controlled, time-varying potential to a *working electrode* immersed in an electrolyte solution and records the resulting current (Fig. 2.1). A reference electrode provides a stable potential baseline and a counter electrode completes the electrical circuit.

The reaction at the electrode surface depends on the applied potential and the species in solution. In the simplest case, a solution-phase species undergoes a single-electron redox reaction:



where A gains one electron (reduction) to form B; and B loses an electron (oxidation) to form A.

2.1 Voltammetry

Voltammetry is the class of techniques in which the potential applied to the working electrode is varied and the resulting current recorded. The current–potential curve, called a *voltammogram*, encodes information about both the reaction kinetics and the transport of species in solution.

In a cyclic voltammetry (CV) experiment the potential is swept linearly from an initial value E_i , where no reaction takes place, to a vertex potential E_v where conversion between species occurs, and then back to E_i . The current I is recorded throughout. An example of the triangular waveform and the resulting voltammogram are shown in Fig. 2.2.

The forward sweep produces a reduction (cathodic) peak; the return sweep produces an oxidation (anodic) peak¹. The peak shape arises from competition between

¹Following IUPAC recommendations for sign conventions of electrochemical data, positive values will be assigned to anodic currents.

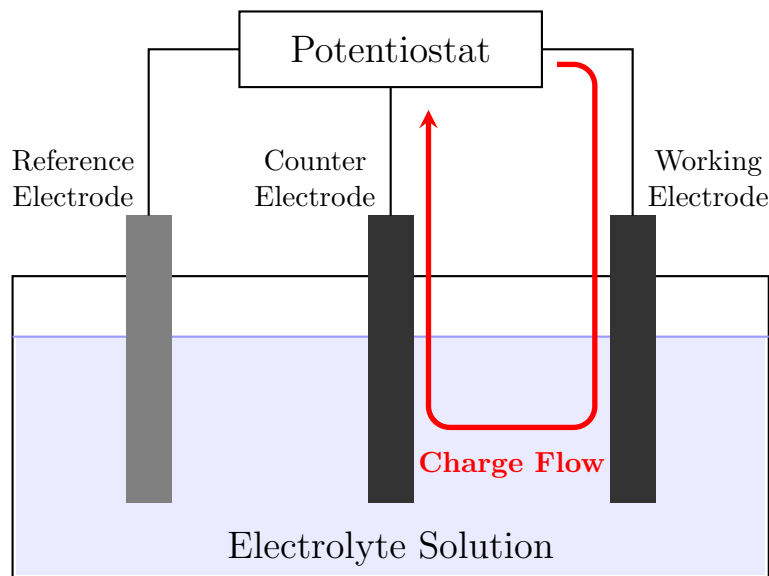


Figure 2.1: Schematic diagram of a three-electrode electrochemical cell. The potentiostat controls the potential difference between the working and reference electrodes whilst driving current (red) through the circuit formed by the working and counter electrodes via the electrolyte solution.

two rate-limiting processes. Early in the forward sweep the driving force for reduction is small and negligible current flows. As the potential moves to more negative values the current increases. The system is now under *kinetic control*, with the rate of electron transfer as the limiting step. Eventually species A near the electrode is depleted and the current peaks. Beyond this point the current decreases despite the applied potential increasing. This is due to the system passing into *diffusional control*: diffusive transport of fresh A to the surface becomes the limiting factor.

The peak positions and separation between anodic and cathodic peaks carry information about the electrode kinetics. The *scan rate* ν (V s^{-1}) controls the timescale of the experiment. Faster scans probe faster kinetic processes, while slower scans allow more time for diffusion and can reveal coupled chemical reactions. The scan rate is chosen for each reaction mechanism so that the current response is sensitive to the target parameters. The rationale for each choice is given when the reaction mechanism is introduced.

2.2 Mathematical Description

We consider a planar disc electrode of radius r_e which is much larger than the diffusion length on the timescale of the experiment. This assumption allows us to model only

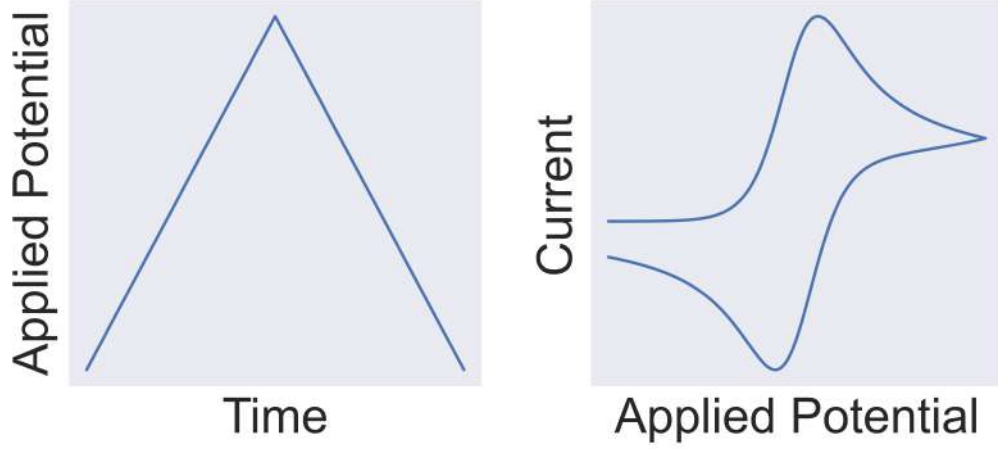


Figure 2.2: Left: the triangular potential waveform applied in a cyclic voltammetry experiment. Right: the resulting voltammogram, showing the cathodic (reduction) peak on the forward sweep and the anodic (oxidation) peak on the reverse sweep.

the diffusion perpendicular to the surface of the electrode reducing the problem to one spatial dimension. (Fig. 2.3).

2.2.1 Diffusion

Transport in unstirred solution is governed by Fick's second law [7]:

$$\frac{\partial c_j}{\partial t} = D_j \frac{\partial^2 c_j}{\partial x^2}, \quad j \in \{A, B\}, \quad (2.2)$$

where D_j ($\text{m}^2 \text{s}^{-1}$) is the diffusion coefficient and x is the distance from the electrode surface. The domain is semi-infinite, $x \geq 0$.

2.2.2 Electrode Kinetics

The diffusive flux of species j at a point x^* is given by Fick's first law [7]:

$$j_{x^*} = -D_j \left(\frac{\partial c_j}{\partial x} \right) \Big|_{x=x^*}. \quad (2.3)$$

The rate of reaction Eq. (2.1) at the electrode surface is governed by the Butler–Volmer equation [9]:

$$j_{0,A} = k_0 \left[c_A(t, 0) e^{-\alpha \frac{F}{\mathcal{RT}} (E(t) - E_f^0)} - c_B(t, 0) e^{(1-\alpha) \frac{F}{\mathcal{RT}} (E(t) - E_f^0)} \right] \quad (2.4)$$

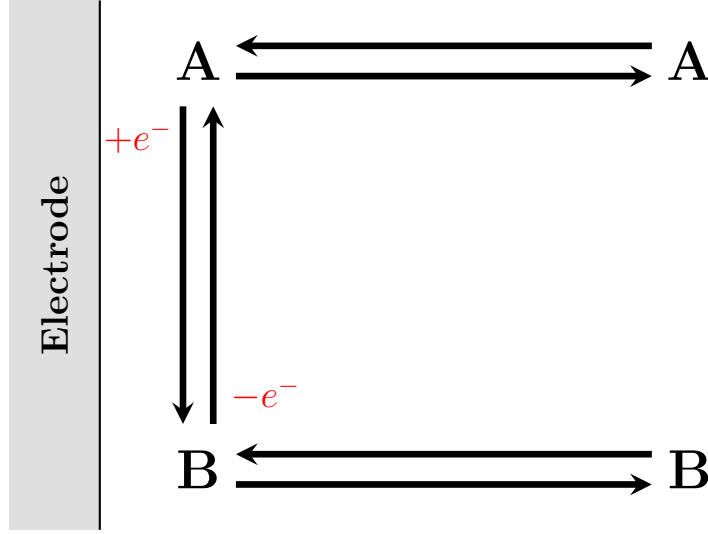


Figure 2.3: Schematic of the one-dimensional electrode reaction. Species A diffuses from the bulk solution towards the electrode surface, where it is reduced ($+e^-$) to form species B. The reverse oxidation ($-e^-$) converts B back to A. Species B diffuses away from the electrode into the bulk. Each processes also happens in reverse.

where k_0 (ms^{-1}) is the standard rate constant, $\alpha \in (0, 1)$ is the transfer coefficient, F is Faraday's constant, $\mathcal{R}$ is the universal gas constant, and $\mathcal{T}$ is the absolute temperature.

Combining Eqs. (2.3) and (2.4) at $x = 0$ gives a Neumann boundary condition:

$$-D_A \left(\frac{\partial c_A}{\partial x} \right) \Big|_{x=0} = k_0 \left[c_A(t, 0) e^{-\alpha \frac{F}{\mathcal{R}\mathcal{T}} (E(t) - E_f^0)} - c_B(t, 0) e^{(1-\alpha) \frac{F}{\mathcal{R}\mathcal{T}} (E(t) - E_f^0)} \right] \quad (2.5)$$

Stoichiometric conservation at the surface requires

$$D_B \left(\frac{\partial c_B}{\partial x} \right) \Big|_{x=0} = -D_A \left(\frac{\partial c_A}{\partial x} \right) \Big|_{x=0}. \quad (2.6)$$

2.2.3 Boundary and Initial Conditions

Initially both species are at their bulk concentrations:

$$c_A(0, x) = c_A^*, \quad c_B(0, x) = c_B^*. \quad (2.7)$$

Far from the electrode the bulk values are recovered:

$$c_j(t, x) \rightarrow c_j^* \quad \text{as } x \rightarrow \infty, \quad j \in \{A, B\}. \quad (2.8)$$

2.2.4 Current

The measured current is

$$I = F A_{\text{el}} j_{0,A} \quad (2.9)$$

where $A_{\text{el}} = \pi r_e^2$ is the electrode area and $j_{0,A}$ is the surface flux of A from Eq. (2.3).

2.3 Non-Dimensionalisation

Equations (2.2) and (2.5) to (2.8) constitute an initial-boundary value problem. We non-dimensionalise by scaling concentrations by c_A^* , diffusion coefficients by D_A , lengths by r_e , and potentials by the thermal voltage $\mathcal{RT}/F$. Table 2.1 summarises the scalings.

Parameter	Normalisation	Parameter	Normalisation
concentration	$C_j = c_j/c_A^*$	formal potential	$\theta_f = (F/\mathcal{RT}) E_f^0$
diffusion coefficient	$d_j = D_j/D_A$	rate constant	$K_0 = k_0 r_e/D_A$
spatial coordinate	$X = x/r_e$	scan rate	$\sigma = (r_e^2/D_A)(F/\mathcal{RT}) \nu$
time	$T = D_A t/r_e^2$	current	$J = I/(\pi r_e F D_A c_A^*)$
applied potential	$\theta = (F/\mathcal{RT}) E$		

Table 2.1: Dimensionless normalisation of variables.

Under these scalings the governing equations become

$$\frac{\partial C_j}{\partial T} = d_j \frac{\partial^2 C_j}{\partial X^2}, \quad (2.10)$$

with initial conditions $C_A(0, X) = 1$, $C_B(0, X) = C_B^*$. The semi-infinite domain is truncated at $X_{\text{max}} = 6\sqrt{T_{\text{max}}}$ and $T_{\text{max}} = 2(\theta_v - \theta_i)/\sigma$. This truncation is chosen so that X_{max} exceeds the diffusion length $\sqrt{2T_{\text{max}}}$ [10] by a wide margin [30], and the far-field conditions $C_A = 1$, $C_B = C_B^*$ are imposed there.

The dimensionless Butler–Volmer boundary condition and mass conservation at $X = 0$ are

$$\begin{aligned} \left. \frac{\partial C_A}{\partial X} \right|_{X=0} &= K_0 [C_A(T, 0) e^{-\alpha(\theta(T) - \theta_f)} - C_B(T, 0) e^{(1-\alpha)(\theta(T) - \theta_f)}], \\ d_B \left. \frac{\partial C_B}{\partial X} \right|_{X=0} &= - \left. \frac{\partial C_A}{\partial X} \right|_{X=0}. \end{aligned} \quad (2.11)$$

The dimensionless current is

$$J_{0,A} = - \left(\frac{\partial C_A}{\partial X} \right) \bigg|_{X=0}, \quad (2.12)$$

so that $I = \pi r_e F D_A c_A^* J_{0,A}$. A single dimensionless simulation applies to any combination of c_A^* , D_A , and r_e .

2.4 Equal Diffusion Coefficients

For the remainder of this dissertation we assume $d_B = 1$. Since both species then satisfy the same diffusion equation with equal and opposite surface fluxes (Eq. (2.11)), the sum $C_A + C_B$ satisfies a diffusion equation with homogeneous Neumann boundary conditions and constant initial and far-field data. It follows that $C_A + C_B = 1 + C_B^*$ everywhere. Writing $C \equiv C_A$, so that $C_B = 1 + C_B^* - C$, the problem reduces to a single PDE with electrode boundary condition

$$\left. \frac{\partial C}{\partial X} \right|_{X=0} = K_0 [C(T, 0) e^{-\alpha(\theta(T) - \theta_f)} - (1 + C_B^* - C(T, 0)) e^{(1-\alpha)(\theta(T) - \theta_f)}]. \quad (2.13)$$

For notational convenience in the chapters that follow, we define

$$\begin{aligned} K_{\text{red}} &= K_0 \exp \left[-\alpha(\theta(T) - \theta_f) \right], \\ K_{\text{ox}} &= K_0 \exp \left[(1 - \alpha)(\theta(T) - \theta_f) \right]. \end{aligned} \quad (2.14)$$

Chapter 3

Numerical Methods

We now solve the initial-boundary value problem from the previous chapter. We focus on the aspects specific to the electrochemical setting: the exponentially expanding spatial mesh and the incorporation of the Butler–Volmer boundary condition into the tridiagonal system.

We set $C_B^* = 0$ throughout, so that the electrode boundary condition Eq. (2.13) becomes

$$\left. \frac{\partial C}{\partial X} \right|_{X=0} = K_{\text{red}}(T) C(T, 0) - K_{\text{ox}}(T) (1 - C(T, 0)), \quad (3.1)$$

where K_{red} and K_{ox} were defined in Eq. (2.14).

3.1 Finite Difference Scheme

Time is discretised uniformly into m steps

$$m = \frac{T_{\text{max}}}{\Delta T} = \frac{2(\theta_v - \theta_i)}{\sigma \Delta T} \quad (3.2)$$

Instead of defining ΔT directly, since this would mean any change in σ would also change the number of temporal steps we use and thus the accuracy of the simulation. We instead define the change in potential per timestep $\Delta\theta = \sigma\Delta T$. This fixes the number of timesteps per solve independent of the scan rate [6].

The concentration gradient is steep near the electrode and negligible in the bulk, so we use an exponentially expanding spatial mesh with initial spacing $h_0 > 0$ and expansion factor $\omega > 1$. The n nodes $\{X_0, \dots, X_{n-1}\}$ are defined by

$$X_i = \begin{cases} \sum_{k=0}^{i-1} h_0 \omega^k, & 0 \leq i \leq n-2, \\ X_{\text{max}}, & i = n-1, \end{cases} \quad (3.3)$$

with $X_0 = 0$ at the electrode surface and local spacings $\Delta X_+ = X_{i+1} - X_i$, $\Delta X_- = X_i - X_{i-1}$. We fix $\omega = 1.1$ throughout.

Applying backward Euler in time and a three-point central difference on the non-uniform mesh to Eq. (2.10) (with $d_j = 1$), the interior nodes $1 \leq i \leq n-2$ satisfy the tridiagonal system

$$\alpha_i C_{i-1}^k + \beta_i C_i^k + \gamma_i C_{i+1}^k = C_i^{k-1}, \quad (3.4)$$

where $C_i^k \approx C(T^k, X_i)$ and the coefficients are

$$\alpha_i = -\frac{2 \Delta T}{\Delta X_- (\Delta X_+ + \Delta X_-)}, \quad \gamma_i = -\frac{2 \Delta T}{\Delta X_+ (\Delta X_+ + \Delta X_-)}, \quad \beta_i = 1 - \alpha_i - \gamma_i. \quad (3.5)$$

At the electrode surface ($i = 0$) a two-point forward difference approximation to Eq. (3.1) gives

$$\frac{C_1^k - C_0^k}{h_0} = K_{\text{red}}^k C_0^k - K_{\text{ox}}^k (1 - C_0^k), \quad (3.6)$$

which rearranges to $\beta_0 C_0^k + \gamma_0 C_1^k = b_0$ with

$$\beta_0 = 1 + h_0(K_{\text{red}}^k + K_{\text{ox}}^k), \quad \gamma_0 = -1, \quad b_0 = h_0 K_{\text{ox}}^k. \quad (3.7)$$

The far-field node imposes $C_{n-1}^k = 1$. The resulting tridiagonal system is solved in $O(n)$ operations by the Thomas algorithm.

3.2 The Forward Map

The solver computes the full concentration field $C(T, X)$, but the experimental observable is the current, which depends only on the surface flux $J_{0,A}$ at each timestep. We therefore define the *forward map*

$$\mathcal{P}: \Phi \rightarrow \mathcal{J}, \quad \phi \mapsto \mathcal{P}(\phi), \quad (3.8)$$

where $\Phi \subseteq \mathbb{R}^3$ is the parameter space with $\phi = (\alpha, K_0, \theta_f)$ and $\mathcal{J}$ is the space of dimensionless flux trajectories evaluated at the discrete time levels. A single evaluation of $\mathcal{P}$ consists of assembling and solving the tridiagonal system at each of the m timesteps and extracting the surface flux. The current is the experimental observable and the sole target of inference, so we evaluate it with a higher-order three-point formula. Fitting a quadratic through (X_0, C_0^k) , (X_1, C_1^k) , (X_2, C_2^k) and differentiating at X_0 gives

$$J_{0,A}^k = -\left(\frac{\partial C}{\partial X}\right)\bigg|_{X=0} \approx -\frac{h_2^2 (C_1^k - C_0^k) - h_1^2 (C_2^k - C_0^k)}{h_1 h_2 (h_2 - h_1)}, \quad (3.9)$$

where $h_1 = X_1 - X_0$ and $h_2 = X_2 - X_0$.

We verify the solver against analytical results in two limiting regimes. In the *reversible* limit ($K_0 \gg \sqrt{\sigma}$), the Randles–Ševčík result [27, 28] gives

$$J_{\text{p,rev}} = -0.4463 \sqrt{\sigma}. \quad (3.10)$$

In the *irreversible* limit ($K_0 \ll \sqrt{\alpha\sigma}$) [6],

$$J_{\text{p,irr}} = -0.4958 \sqrt{\alpha\sigma}, \quad \theta_{\text{p,irr}} = \theta_f + \frac{1}{\alpha} \left[\ln \left(\frac{K_0}{\sqrt{\alpha\sigma}} \right) - 0.78 \right]. \quad (3.11)$$

Figure 3.1 compares the numerical flux with both expressions; the relative error in the peak current is below 0.1% in each case. These mesh and timestep parameters are reused for all electron-transfer experiments in subsequent chapters.

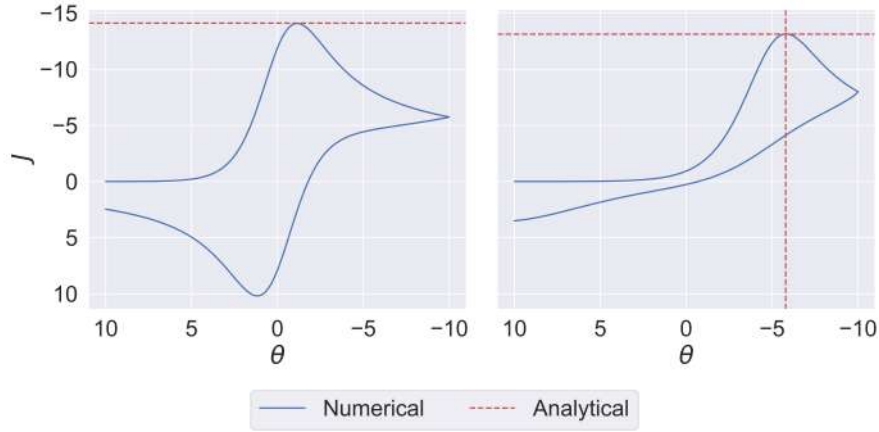


Figure 3.1: Verification of the forward solver with $\sigma = 1000$. Left: reversible limit ($K_0 = 1000$, $\alpha = 0.5$), compared with the Randles–Ševčík peak current Eq. (3.10). Right: irreversible limit ($K_0 = 1$, $\alpha = 0.5$), compared with Eq. (3.11).

3.3 Convergence Study

The discretisation parameters were selected by refinement against a reference solution computed on a fine mesh ($h_0 = 10^{-10}$, $\Delta\theta = 10^{-5}$). We swept h_0 and $\Delta\theta$ independently, measuring the relative L^2 error of the current against the reference. The spatial error plateaus for $h_0 \lesssim 10^{-6}$. The temporal error converges at the expected first-order rate for the backward Euler scheme. We adopt $h_0 = 10^{-6}$ and $\Delta\theta = 0.004$ throughout (Fig. 3.2), which places the solution well within the converged regime at acceptable runtime. The same methodology was applied to the heterogeneous and adsorption solvers (Appendix A).

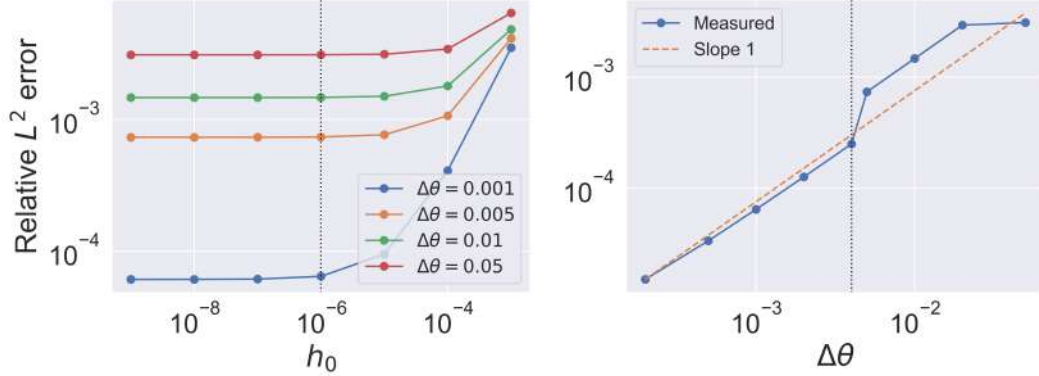


Figure 3.2: Convergence of the electron-transfer solver. Relative L^2 error of the simulated current against a reference solution ($h_0 = 10^{-10}$, $\Delta\theta = 10^{-5}$). Left: spatial refinement at four fixed $\Delta\theta$; the error plateaus for $h_0 \lesssim 10^{-6}$ where the temporal error dominates. Right: temporal refinement at $h_0 = 10^{-6}$; the measured slope matches the first-order rate expected for backward Euler. Dotted lines mark the adopted values $h_0 = 10^{-6}$ and $\Delta\theta = 4 \times 10^{-3}$.

3.4 Effect of Butler–Volmer Parameters

The behaviour of the voltammogram depends on the *kinetic regime*, which is characterised by the dimensionless ratio

$$\Lambda = \frac{K_0}{\sqrt{\sigma}}. \quad (3.12)$$

When $\Lambda \gg 1$ the reaction is reversible and the electron transfer is fast relative to diffusion; when $\Lambda \ll 1$ it is irreversible. The intermediate quasi-reversible regime, where both kinetics and diffusion influence the voltammogram, is the most informative for parameter estimation.

Figure 3.3 illustrates the effect of each Butler–Volmer parameter in the quasi-reversible regime. The transfer coefficient α controls the asymmetry between cathodic and anodic peaks; the rate constant K_0 governs peak sharpness and separation; the formal potential θ_f translates the voltammogram along the potential axis without altering peak shapes. Because changes in one parameter can partially mimic changes in another – for instance, a smaller K_0 broadens and separates the peaks, which an adjustment in θ_f can partially compensate – the parameters are correlated and therefore more difficult to identify.

The problem is compounded in the reversible regime. Figure 3.4 shows that at $K_0 = 200$ the voltammograms for $\alpha = 0.3, 0.5, 0.7$ are nearly indistinguishable, and

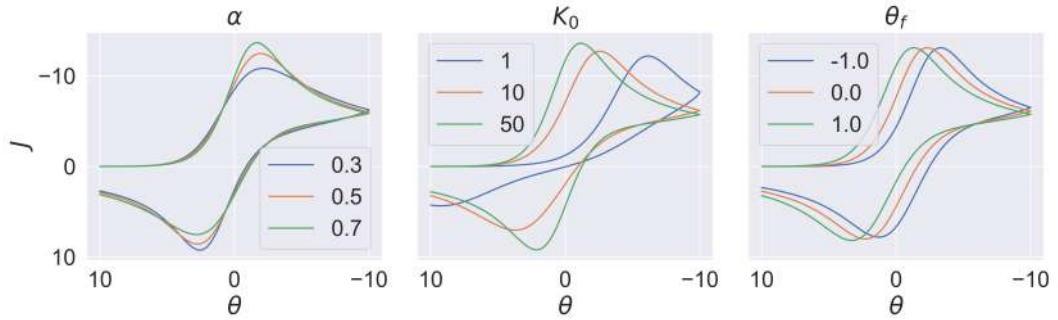


Figure 3.3: Effect of each Butler–Volmer parameter on the dimensionless voltammogram for a quasi-reversible reaction ($K_0 = 20$, $\alpha = 0.6$, $\sigma = 1000$). In each panel one parameter is varied while the others are held fixed. Left: varying α changes the relative heights of the cathodic and anodic peaks. Centre: increasing K_0 symmetrises and sharpens the voltammogram. Right: varying θ_f translates the peaks along the potential axis.

similarly for $K_0 = 200, 500, 1000$. The combination of parameter correlations in the quasi-reversible regime and negligible differences in the reversible regime motivates a probabilistic treatment that can quantify these uncertainties and reveal the correlation structure. We develop this in the next chapter.

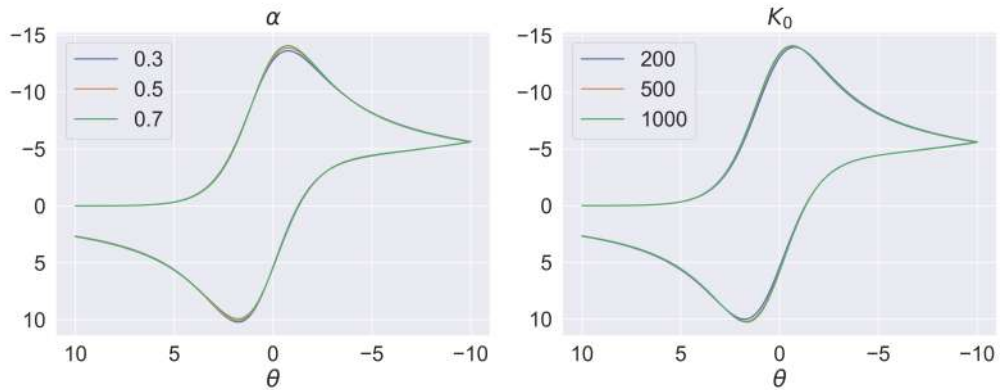


Figure 3.4: Left: effect of α in the reversible regime ($K_0 = 200$); the three curves are nearly indistinguishable. Right: effect of K_0 for values well into the reversible regime; again the voltammograms overlap.

Chapter 4

Bayesian Approach to the Inverse Problem

The correlations and negligible impact of parameters discussed in Section 3.4 make the inverse problem ill-posed: a unique, stable solution may not exist. We therefore take a Bayesian approach and seek probability distributions over the parameters rather than point estimates.

4.1 Noise Model

We model the observed dimensionless flux as the forward map corrupted by Gaussian noise:

$$y_i = \mathcal{P}(\phi)_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \varsigma^2), \quad (4.1)$$

where $\mathbf{y} = (y_1, \dots, y_N)$ are the N flux measurements and $\phi = (\alpha, K_0, \theta_f)$. The noise standard deviation is set as a fraction of the peak flux magnitude:

$$\varsigma = \eta |J_{\text{peak}}|, \quad J_{\text{peak}} = \min_i \mathcal{P}(\phi^*)_i, \quad (4.2)$$

where $\eta \in (0, 1)$ is a dimensionless noise parameter and ϕ^* is the true parameter vector. Experimental measurements typically exhibit noise levels of 1–2% of the peak current [24], corresponding to $\eta \in [0.01, 0.02]$. For a given η , the noise level ς is determined by the noiseless forward solve and is treated as a known constant throughout the inference. Examples of synthetic noisy data are shown in Fig. 4.1.

4.2 Posterior Distribution

Bayes' theorem gives

$$p(\phi | \mathbf{y}) = \frac{L(\mathbf{y} | \phi) \pi(\phi)}{p(\mathbf{y})}, \quad (4.3)$$

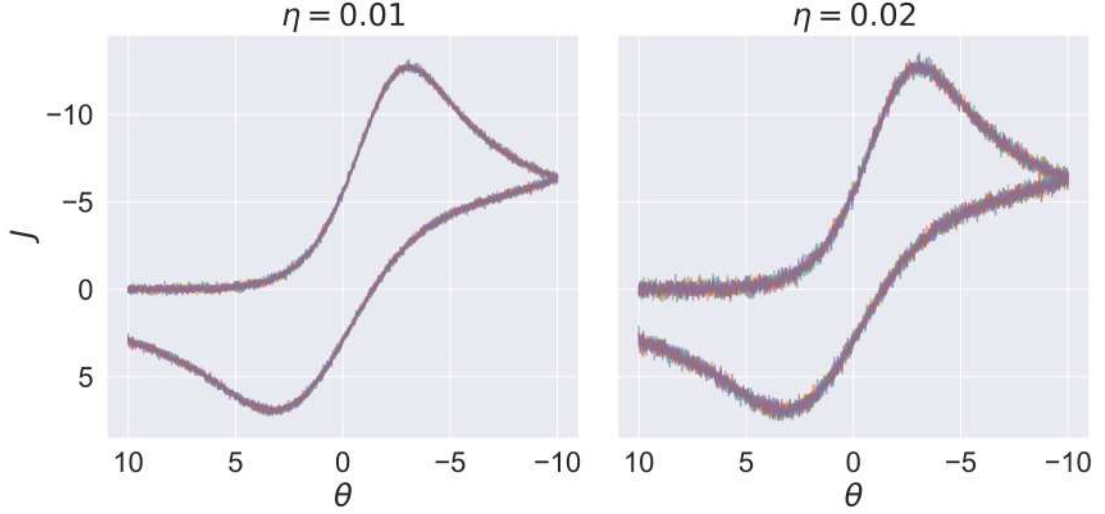


Figure 4.1: Synthetic voltammogram data generated by evaluating the forward map $\mathcal{P}$ at $\phi^* = (0.6, 10.0, 0.5)$ with dimensionless scan rate $\sigma = 1000$ over the potential window $\theta_i = 10$ to $\theta_v = -10$, then adding i.i.d. Gaussian noise. The noisy observations are the data $\mathbf{y}$ from which the Butler–Volmer parameters are to be inferred.

where $L(\mathbf{y} \mid \phi)$ is the likelihood, $\pi(\phi)$ is the prior, and $p(\mathbf{y})$ is the evidence. We adopt a uniform prior over physically reasonable bounds ($\alpha \in (0, 1)$, $K_0 > 0$), so that

$$p(\phi \mid \mathbf{y}) \propto L(\mathbf{y} \mid \phi). \quad (4.4)$$

Under the noise model Eq. (4.1) the observations are conditionally independent given ϕ . Dropping constant terms that do not depend on ϕ , the log-likelihood is

$$\ell(\phi) = -\frac{1}{2\zeta^2} \sum_{i=1}^N (y_i - \mathcal{P}(\phi)_i)^2. \quad (4.5)$$

4.3 Random Walk Metropolis–Hastings

The posterior Eq. (4.4) is known only up to a normalising constant. Markov chain Monte Carlo (MCMC) addresses this by constructing a chain whose stationary distribution is the posterior.

In the random walk Metropolis–Hastings (RWMH) algorithm [18, 4], proposals are drawn from a symmetric Gaussian centred at the current state:

$$\phi_{\text{cand}} \sim \mathcal{N}(\phi^{(t)}, \Sigma). \quad (4.6)$$

Proposals are accepted or rejected according to the Metropolis–Hastings acceptance probability [18]

$$r = \min\{\exp(\ell(\phi_{\text{cand}}) - \ell(\phi^{(t)})), 1\}. \quad (4.7)$$

The proposal covariance Σ is tuned to achieve an acceptance rate of approximately 23%, which is asymptotically optimal for Gaussian targets [13].

4.3.1 Sampling Results

We apply RWMH to synthetic data generated with true parameters $\phi^* = (0.6, 10.0, 0.5)$ at scan rate $\sigma = 1000$ and noise level $\eta = 0.02$. Figure 4.2 shows the marginal posterior histograms which confirm all three distributions concentrate around the true values (dashed lines).

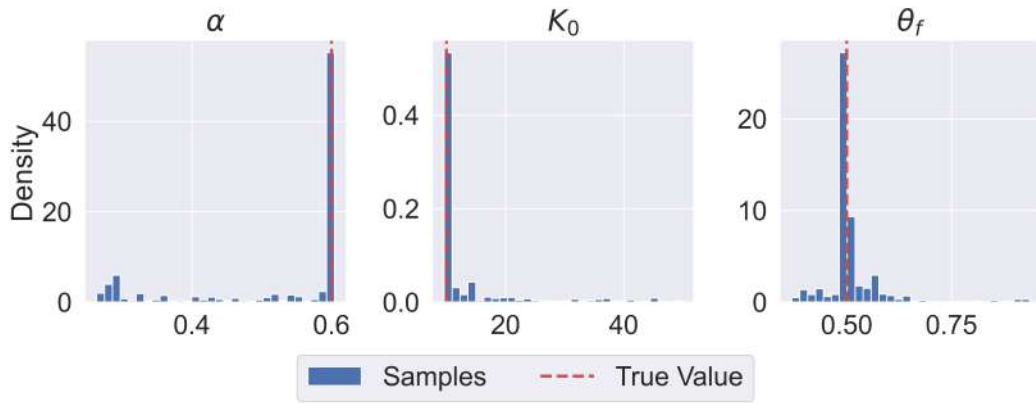


Figure 4.2: Marginal posterior histograms for the three Butler–Volmer parameters obtained by RWMH sampling on synthetic data with noise level $\eta = 0.02$. Red dashed lines indicate the true parameter values used to generate the data.

The histograms exhibit extended tails originating from the chain’s early samples, visible in Fig. 4.3. The chain spends many iterations traversing low-probability areas before settling near the true values. This is due to it being initialised far from the high-density region. These initial samples are typically discarded as *burn-in*. But since each likelihood evaluation requires a full forward solve, this burn-in represents a significant waste of computation.

A gradient of the log-likelihood would address this. We could locate the high-density region by optimisation before sampling begins. Gradient-free optimisers could do the same, but gradient-based methods offer faster and more robust convergence. Moreover, gradients can be incorporated into the MCMC proposal itself, yielding samplers that explore the posterior more efficiently than random walks. Both uses

require differentiating through the forward map $\mathcal{P}$, which we develop in the following chapter.

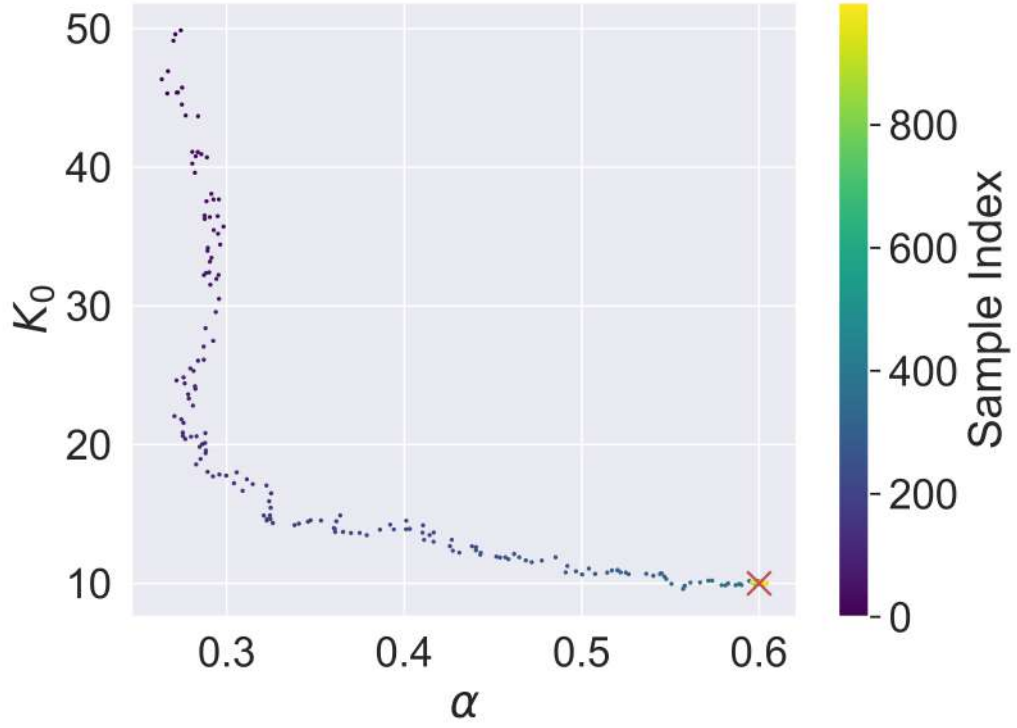


Figure 4.3: Scatter plot of the RWMH chain projected onto the (α, K_0) plane. Points are coloured by sample index: early samples (dark) show the chain traversing a low-density region far from the true parameters (red cross), while later samples (light) concentrate near the true values.

Chapter 5

Differentiating Through the Forward Solver

Computing the gradient $\nabla_{\phi}\ell$ of the log-likelihood Eq. (4.5) requires differentiating through the forward map $\mathcal{P}$, which involves solving a sequence of tridiagonal linear systems. In this chapter we derive an efficient adjoint method for this computation and demonstrate its value through gradient-based optimisation.

5.1 Discretise Then Optimise

There are two main strategies for differentiating through a numerical solver. The *optimise then discretise* approach derives continuous adjoint equations and then discretises them; the *discretise then optimise* approach differentiates the discrete computation directly. We adopt the latter, which yields the exact gradient of the quantity evaluated in code and avoids consistency errors between the forward and adjoint discretisations [20].

5.2 Reverse-Mode AD and the Cost of Naïve Taping

Since ℓ is scalar-valued, reverse-mode automatic differentiation (AD) computes the full gradient in a single backward pass. It decomposes the computation into a sequence of *vector-Jacobian products* (VJPs).

For $f = f_L \circ \cdots \circ f_1$, the chain rule gives

$$\frac{\partial f}{\partial \phi} = J_{f_L} J_{f_{L-1}} \cdots J_{f_1}, \quad (5.1)$$

which reverse mode AD evaluates from left to right, multiplying the cotangent vector by each Jacobian in turn without forming the intermediate matrices explicitly. In the present setting $f = \mathcal{P}$.

The dominant operations in $\mathcal{P}$ are the m tridiagonal solves, so the central question is how to implement the VJP of a linear solve $Ax = b$. A generic AD framework will tape every intermediate value of the Thomas algorithm – elimination multipliers, modified diagonals, partial sums – and replay them in reverse. The overhead of maintaining and traversing the tape within the linear solve can degrade runtime performance significantly.

5.3 Adjoint of the Tridiagonal Solve

We avoid taping by exploiting the structure of the linear solve. Given the forward system $Ax = b$ and a scalar loss ℓ , the task is to express the adjoints $\bar{b} = \partial\ell/\partial b$ and $\bar{A} = \partial\ell/\partial A$ in terms of the incoming cotangent $\bar{x} = \partial\ell/\partial x$.

Define λ by the *adjoint system*

$$A^T \lambda = \bar{x}. \quad (5.2)$$

Differentiating $Ax = b$ and allowing both A and b to vary gives

$$dA x + A dx = db, \quad (5.3)$$

Left-multiplying Eq. (5.3) by λ^T and applying Eq. (5.2)

$$\lambda^T dA x + \bar{x}^T dx = \lambda^T db, \quad (5.4)$$

by rearranging and noting $d\ell = \bar{x}^T dx$

$$d\ell = \bar{x}^T dx = \lambda^T db - \lambda^T dA x. \quad (5.5)$$

Reading off the coefficients of db and dA identifies the adjoints:

$$\bar{b} = \lambda^T, \quad \bar{A} = -\lambda x^T. \quad (5.6)$$

Two observations make this practical. First, A^T is tridiagonal, so the adjoint system Eq. (5.2) is solved by the Thomas algorithm at the same $O(n)$ cost as the forward solve. Second, $\bar{A}$ is never formed as a $n \times n$ matrix: only the three diagonals are needed, and $\bar{A}_{ij} = -\lambda_i x_j$ on each diagonal reduces to the elementwise products

$$\bar{a}_i = -\lambda_{i+1} x_i, \quad \bar{b}_i^{\text{diag}} = -\lambda_i x_i, \quad \bar{c}_i = -\lambda_i x_{i+1}, \quad (5.7)$$

for the sub-, main- and super-diagonals of A .

Appendix C lists the JAX [2] implementation. The solver takes the three diagonals a, b, c and right-hand side d as separate arrays. The forward pass returns x and caches (a, b, c, x) . The backward pass solves the adjoint system by calling the same Thomas routine with the sub- and super-diagonals swapped and returns the three diagonal adjoints from Eq. (5.7) together with $\bar{d} = \lambda$.

5.3.1 Full Backward Pass

The custom VJP handles a single timestep. The full gradient $\nabla_{\phi}\ell$ is assembled by walking backward through time levels $k = m, m - 1, \dots, 1$. At each level the adjoint system Eq. (5.2) is solved using the cached A^k diagonals and x^k , yielding λ^k and the matrix adjoint $\bar{A}^k = -\lambda^k(x^k)^T$. Only the first row of A^k depends on ϕ through K_{red} and K_{ox} . The interior rows are parameter-independent, so the contributions to $\nabla_{\phi}\ell$ come exclusively from $\bar{A}^k$ applied to the partial derivatives $\partial A_{0\cdot}^k / \partial \phi$. The result is the exact gradient of the discrete log-likelihood at a cost of one adjoint solve per timestep.

The asymptotic complexity of the backward pass is unchanged: both naïve taping and the adjoint require $O(mn)$ work, since each of the m timesteps needs one $O(n)$ backward solve. The saving is in the memory constant and in eliminating tape-traversal overhead. For the electron reaction this reduces the backward pass from 15.8 ms with standard AD to 7.25 ms with the custom rule.

5.4 Extension to Pentadiagonal Systems

The adjoint generalises to pentadiagonal systems: the transpose of such a matrix retains the same band structure, so the adjoint system is solved at the same $O(n)$ cost per timestep. We exploit this in Chapters 7 and 8, where coupled multi-species reactions lead to pentadiagonal structure. The saving grows with bandwidth. For the heterogeneous reaction of Chapter 7 the backward pass takes 43.7 ms with the custom VJP against 481 ms for standard AD. This more than 10 $\times$ reduction in the backward pass makes scaling to wider-band problems practical within our runtime budget.

5.5 Optimisation Results

We now use the adjoint gradient to locate the high-density region of the posterior and compare the gradient-based ADAM optimiser [8, 21] against CMA-ES [17, 23], a gradient-free evolutionary strategy that adapts a population covariance over generations, used in Gavaghan et al. [11].

A single ADAM step requires one forward and one adjoint solve, costing approximately two forward-solve equivalents. We use the recommended CMA-ES population size of $\lambda = 4 + \lfloor 3 \ln d \rfloor = 7$ [17], requiring λ forward solves per generation. We equalise the total budget in forward-solve equivalents so that both methods consume the same runtime.

Figure 5.1 shows the log-posterior density over the course of optimisation at two noise levels ($\eta = 0.01$ and $\eta = 0.02$), each run from 32 initial conditions drawn by Latin hypercube sampling centred at the true parameters. ADAM reaches the high-density threshold (dashed line) in fewer forward-solve equivalents and with lower variance across initialisations. In three dimensions both methods reach the region reliably, so the gradient advantage is marginal; it becomes important in higher dimensions (Appendix B).

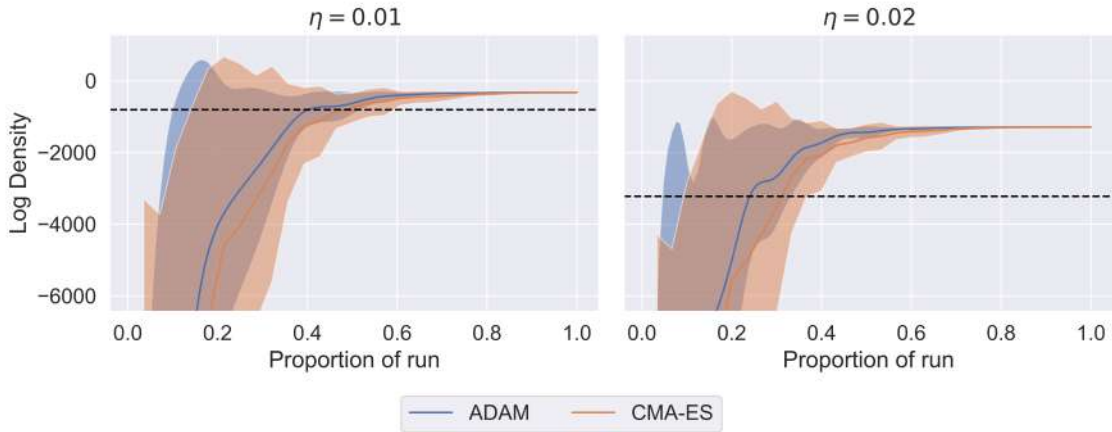


Figure 5.1: Log-posterior density during optimisation for ADAM and CMA-ES (population size 7) at noise levels $\eta = 0.02$ and $\eta = 0.01$. Each algorithm is run from 32 randomly drawn initial conditions; solid lines show the mean and shaded regions indicate ± 1 standard deviation across runs. The black dashed line marks the two times the log-density evaluated at the true parameters ϕ^* – indicating high density threshold. Both algorithms are given equal total computational budgets measured in forward-solve equivalents. 105 steps for ADAM and 30 iterations for CMA-ES.

Chapter 6

Hamiltonian Monte Carlo

We can generate proposals guided by the posterior geometry with the adjoint gradients now available. *Hamiltonian Monte Carlo* (HMC) [1, 25] does this by borrowing the formalism of Hamiltonian dynamics: the current state of the Markov chain is treated as a particle that receives a random momentum kick and then evolves under a Hamiltonian coupling its position to the posterior. Because the dynamics conserve energy, the particle traces out long-range trajectories through regions of high posterior density that would take a random walk many steps to traverse.

6.1 The Hamiltonian

Let $q \in \mathbb{R}^d$ denote the position (the parameter vector ϕ ; we use q to follow the Hamiltonian mechanics convention) and $p \in \mathbb{R}^d$ an auxiliary momentum. We define the Hamiltonian

$$H(q, p) = U(q) + K(p), \quad (6.1)$$

where the potential energy $U(q) = -\ell(q) + \text{const}$ is the negative log-posterior and the kinetic energy is

$$K(p) = \frac{1}{2} p^T M^{-1} p. \quad (6.2)$$

The mass matrix $M \in \mathbb{R}^{d \times d}$ determines the momentum distribution $p \sim \mathcal{N}(0, M)$; choosing M to approximate the posterior covariance preconditions the dynamics so that the particle explores all directions at a similar rate.

6.2 Leapfrog Integration and Metropolis Correction

Hamilton's equations

$$\frac{dq}{dt} = M^{-1}p, \quad \frac{dp}{dt} = -\nabla_q U(q) = \nabla_q \ell(q) \quad (6.3)$$

are integrated numerically using the *leapfrog* integrator, which is symplectic and time-reversible. Given step size ε and L steps, one trajectory proceeds as

$$\begin{aligned} p_{i+1/2} &= p_i - \frac{\varepsilon}{2} \nabla_q U(q_i), \\ q_{i+1} &= q_i + \varepsilon M^{-1} p_{i+1/2}, \\ p_{i+1} &= p_{i+1/2} - \frac{\varepsilon}{2} \nabla_q U(q_{i+1}), \end{aligned} \quad (6.4)$$

for $i = 0, \dots, L-1$. Each step requires both a forward and adjoint solve, with the second gradient reused in the next step. Because the integrator is symplectic, the energy error along the trajectory is small and the endpoint (q^*, p^*) is accepted with probability

$$r = \min\{1, \exp(H(q, p) - H(q^*, p^*))\}, \quad (6.5)$$

giving high acceptance rates even for distant proposals.

6.3 The No-U-Turn Sampler

The efficiency of HMC depends on the trajectory length L : too short and the sampler reverts to near-random-walk behaviour; too long and the trajectory doubles back, wasting gradient evaluations. The *No-U-Turn Sampler* (NUTS) [4, 19] eliminates L as a tuning parameter by doubling the trajectory in both directions until the endpoints begin to converge (a U-turn), then drawing a sample uniformly from the valid states. This leaves only ε and M to be specified.

6.4 Parameter Transformations

The leapfrog integrator applies a uniform step size in every coordinate, so parameters with bounded support or disparate scales require transformation to an unconstrained space. We apply the logit transform to the transfer coefficient and the log transform to the rate constant:

$$\tilde{\alpha} = \text{logit}(\alpha) = \log\left(\frac{\alpha}{1-\alpha}\right), \quad \tilde{K}_0 = \log K_0. \quad (6.6)$$

The formal potential θ_f is already unconstrained. The sampler operates in $\tilde{\phi} = (\tilde{\alpha}, \tilde{K}_0, \theta_f)$ and the transformations are composed with the forward map so that reverse-mode AD propagates gradients through them automatically.

6.5 Inference Workflow

The following four-step pipeline is used for all reactions in this dissertation. All runtimes were measured on an Apple M1 with eight cores; no GPU acceleration was used.

Step 1: Diverse initialisation. $N_{\text{init}} = 8$ starting points are drawn in the transformed space via Latin hypercube sampling to ensure broad prior coverage.

Step 2: Gradient-based optimisation to the high-density region. Each start is optimised independently with ADAM for a fixed number of steps. The step budget is chosen to move iterates into the high-density region of the posterior without collapsing them onto a single point.

Step 3: Window adaptation. Window adaptation [29, 4] is run once, from one of the ADAM endpoints, to tune the leapfrog step size ε via dual averaging and to estimate the mass matrix M from the warm-up sample covariance. The tuned (ε, M) are shared across all eight NUTS chains. The RWMH proposal covariance is set to $\Sigma = (2.38^2/d) M^{-1}$, the asymptotically optimal scaling for Gaussian targets [13].

Step 4: Equal-budget parallel sampling. NUTS and RWMH chains are run in parallel from the eight ADAM endpoints. Each NUTS iteration costs L leapfrog steps; each RWMH iteration costs one forward solve. At a matched sampling wall-time of approximately 80 s, we obtain 2,400 NUTS samples and 24,000 RWMH samples in total.

6.6 Results

6.6.1 Posterior Distributions

Both samplers recover good posteriors across the regimes tested below. This is due to the optimisation stage of Section 6.5 placing every chain inside the high-density region before sampling begins.

Figure 6.1 shows the joint and marginal posteriors from NUTS and RWMH for the quasi-reversible electron-transfer reaction with $\phi^* = (0.6, 10.0, 0.5)$ at $\sigma = 1000$ ($\Lambda = K_0/\sqrt{\sigma} \approx 0.32$). Both samplers recover the true values, with tight marginals in close agreement. The off-diagonal panels show weak pairwise dependence between all three parameters, with a mild negative correlation between K_0 and θ_f .

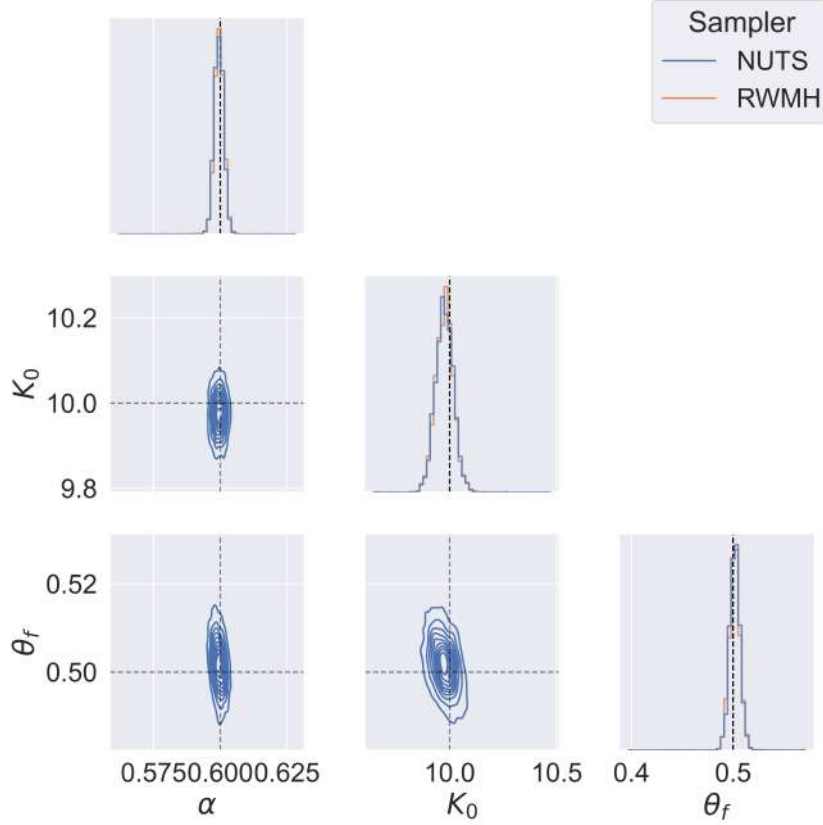


Figure 6.1: Corner plot of the posterior from NUTS (blue) and RWMH (orange) under equal computational budgets for the quasi-reversible case ($\phi^* = (0.6, 10.0, 0.5)$, $\sigma = 1000$). Diagonal: marginal histograms for both samplers. Lower triangle: NUTS joint density contours. Dashed lines indicate the true values.

We repeat the experiment in the reversible regime with $\phi^* = (0.6, 500.0, 0.5)$, giving $\Lambda \approx 15.8 \gg 1$. Figure 6.2 shows that the α and K_0 marginals are now wide compared to the quasi-reversible regimes. While the formal potential θ_f remains tightly constrained and largely independent of the other two. Both samplers remain centred very close to the true values.

Both the correlation in Figure 6.1 and marginal widths in Fig. 6.2 are precisely the structure that motivated the Bayesian treatment in Chapter 4. A point estimate

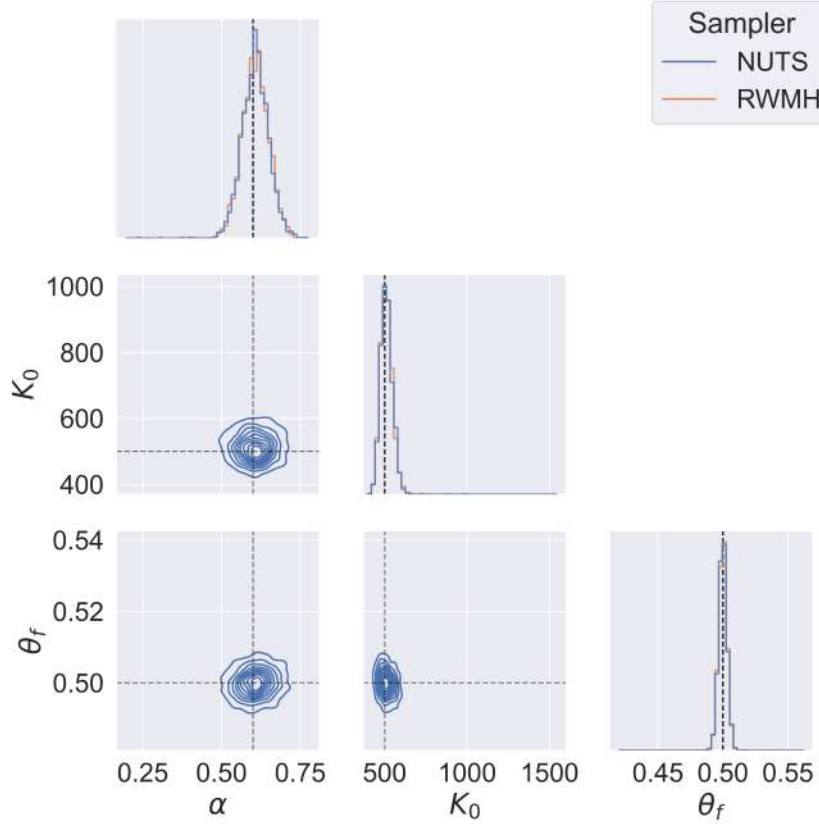


Figure 6.2: Corner plot in the reversible regime ($\phi^* = (0.6, 500.0, 0.5)$, $\Lambda \approx 15.8$). The α and K_0 marginals are wider than in Fig. 6.1, reflecting the insensitivity discussed in Section 3.4. The formal potential θ_f remains tightly constrained.

would report a single (α, K_0) pair in the reversible regime and miss the range of equally plausible values, and would not show the K_0 – θ_f dependence in the quasi-reversible case.

Figure 6.3 shows the posterior-mean voltammogram and 95% credible interval for both regimes. The credible interval on the current is tight in both cases despite the wide (α, K_0) marginals in the reversible regime. This indicates that many combinations of α and K_0 values produce similar voltammograms in this regime.

6.6.2 Diagnostics

Both samplers produce marginals centred on the true values with similar widths, but this does not establish that the two methods are equally reliable. Proposals from MCMC methods depend on their current state and can therefore be correlated. This is an issue since correlated samples contain less information than independent ones.

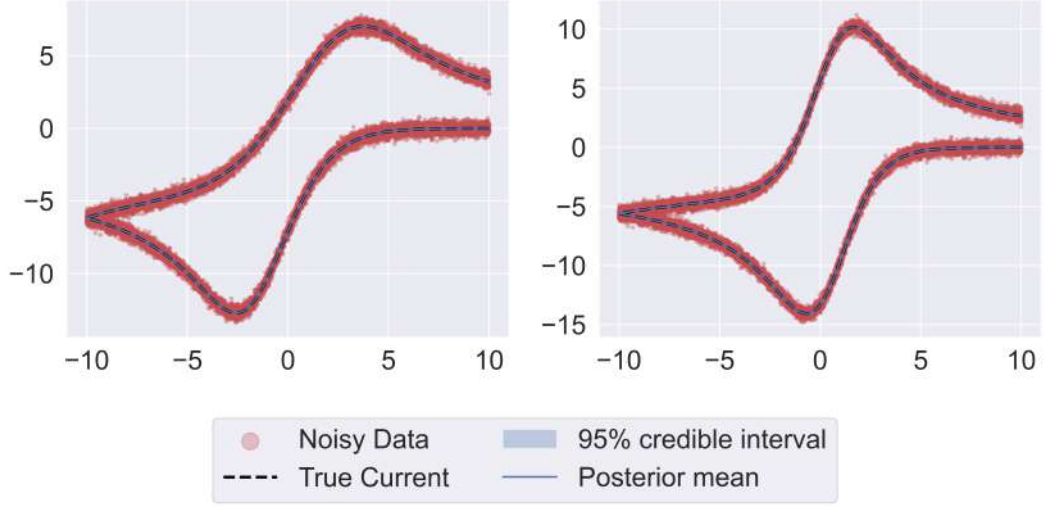


Figure 6.3: Posterior-mean voltammogram (solid) with 95% credible interval (shaded) of the NUTS posterior against the true current (dashed) and noisy data (points). Left: quasi-reversible regime. Right: reversible regime. The credible interval remains tight in both cases despite the wider parameter posteriors in the reversible regime.

Furthermore, we seek some measure of how stable the posterior distributions are: if we stopped sampling now, would the reported moments be the same as if we had run for longer?

Effective Sample Size (ESS) quantifies the number of independent draws equivalent to the correlated chain [12, 15]:

$$\text{ESS} = \frac{T}{1 + 2 \sum_{k=1}^{\infty} \rho_k}, \quad (6.7)$$

where T is the chain length and ρ_k the autocorrelation at lag k .

When the autocorrelations are predominantly positive, $\text{ESS} < T$. A sampler can also produce *antithetic* draws, in which case the denominator falls below one and ESS exceeds T [29, 31]. NUTS can exhibit this behaviour for parameters whose marginal posterior is close to Gaussian with weak dependence on the remaining parameters [29]. Table 6.1 compares the ESS for NUTS and RWMH. NUTS achieves $1.6\text{--}1.9\times$ the ESS of RWMH across the parameters. For K_0 , the NUTS ESS exceeds the chain length ($2834.3 > 2400$) meaning the samples are antithetic.

Gelman–Rubin Statistic ($\hat{R}$) assesses whether chains initialised from dispersed starting points have forgotten those starting points and agree on the posterior. It is

	α	K_0	θ_f
NUTS	1920.7	2834.3	2337.4
RWMH	1211.7	1718.3	1236.7
Ratio	1.6	1.6	1.9

Table 6.1: Effective sample size for each parameter from 24,000 RWMH samples and 2,400 NUTS samples, with NUTS/RWMH ratio.

the reason for running multiple chains in parallel initialised from a different optimised point in Step 4 of Section 6.5. Given M chains of length N , let W denote the mean within-chain variance and B the between-chain variance of the chain means. The statistic [14] is

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \quad \hat{V} = \frac{N-1}{N} W + \frac{1}{N} B, \quad (6.8)$$

where $\hat{V}$ is the pooled estimator of the posterior variance. When chains have not mixed, $B \gg W$ and $\hat{R} \gg 1$ and on mixing, $\hat{R} \rightarrow 1$. The convention $\hat{R} < 1.01$ [31] is adopted as the threshold below which between-chain disagreement is not distinguishable from within-chain Monte Carlo variation. While $\hat{R} > 1.01$, reported moments are biased toward the chains' starting positions and thus not reliable.

Figure 6.4 shows this link for K_0 : the NUTS chains agree from the first checkpoint with $\hat{R} < 1.01$ throughout, while the RWMH inter-chain initial band is wide associated with $\hat{R}$ at 1.2. The chains then tighten their bands over the samples with $\hat{R}$ crossing the threshold at roughly 0.3 of the sample proportion.

Figure 6.5 shows that each parameter in the electron reaction follows a similar pattern: NUTS at $\hat{R} \leq 1.01$ from the first checkpoint, RWMH crossing the threshold at roughly 30% of the samples for α , K_0 , and θ_f . The long-range NUTS proposals, preconditioned by the window-adapted mass matrix, leave the initial state after a handful of iterations. RWMH's local steps retain memory of their starting points for much longer.

The NUTS curves in Fig. 6.5 dip below 1 early in the run before settling at it. NUTS mixes essentially immediately, so the eight chain means scatter around the posterior mean from the first checkpoint, giving $\mathbb{E}[B] \approx \varsigma^2 \approx \mathbb{E}[W]$. In Eq. (6.8), however, W averages eight per-chain variances while B is built from the variance of only eight chain means, with $M-1 = 7$ degrees of freedom. The sampling distribution of B is therefore right-skewed with median below its mean, so the B/W ratio is typically less than one and pulls $\hat{R}$ below 1. The dip is thus a small-sample property of the estimator under immediate mixing rather than evidence of non-convergence.

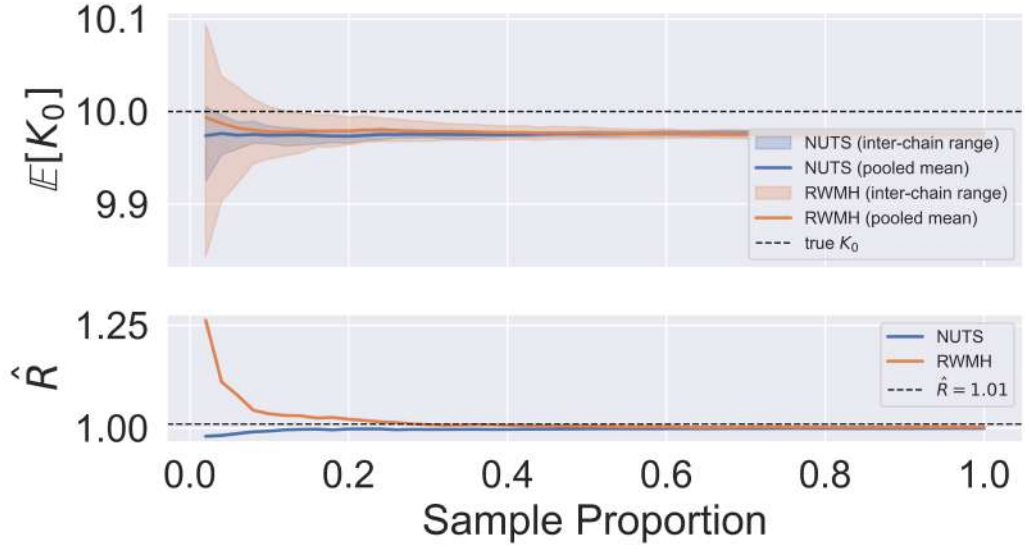


Figure 6.4: Upper: pooled running estimate of $\mathbb{E}[K_0]$ for NUTS (blue) and RWMH (orange) against sample proportion. The shaded band shows the range of per-chain means across the eight chains, which is the between-chain dispersion entering B in Eq. (6.8). Lower: $\hat{R}$ over the same budget. The RWMH running estimate stabilises as $\hat{R}$ crosses the 1.01 threshold. The dashed line marks the $\hat{R} = 1.01$ convergence threshold [31].

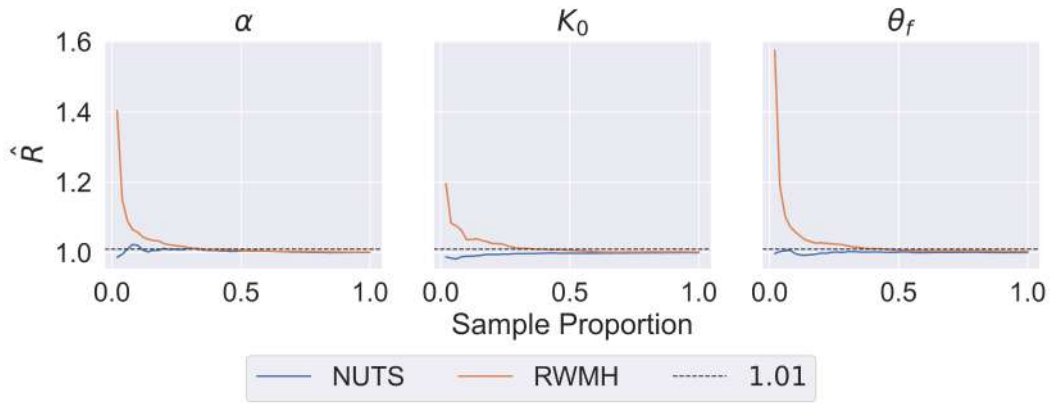


Figure 6.5: $\hat{R}$ as a function of sample proportion for NUTS (blue) and RWMH (orange) for each parameter of the quasi-reversible electron-transfer reaction. NUTS remains below the 1.01. RWMH requires roughly 30% of the samples to converge. The dashed line marks the $\hat{R} = 1.01$ convergence threshold [31].

Chapter 7

Heterogeneous Reaction

This chapter applies the inference framework to a reaction involving two electron transfers coupled by an irreversible heterogeneous chemical step. The parameter space grows from three to seven dimensions. Because the methodology has been established, we focus on what is new: the reaction scheme, the coupled PDE system, the reordering that yields a pentadiagonal linear system, and the inference results. A more detailed treatment of surface processes with explicit surface concentrations and adsorption/desorption kinetics is covered in Chapter 8.

7.1 Reaction Scheme

We consider the $\text{EC}_{\text{irr,het}}\text{E}$ mechanism [6]:



Species A is reduced to B; B is converted irreversibly to C by a surface-catalysed process with rate constant k_{het} (m s^{-1}). C undergoes a second electron transfer to form D. The chemical step occurs only at the electrode surface, so bulk transport of all four species is governed by Fick's second law. This mechanism arises, for example, in the electroreduction of halonitroaromatic compounds in aprotic media [6].

Each electron transfer has its own $\alpha^{(i)}$, $K_0^{(i)}$, and $\theta_f^{(i)}$. Together with the dimensionless heterogeneous rate constant K_{het} , the parameter vector is

$$\phi = (\alpha^{(1)}, K_0^{(1)}, \theta_f^{(1)}, \alpha^{(2)}, K_0^{(2)}, \theta_f^{(2)}, K_{\text{het}}) \in \mathbb{R}^7. \tag{7.2}$$

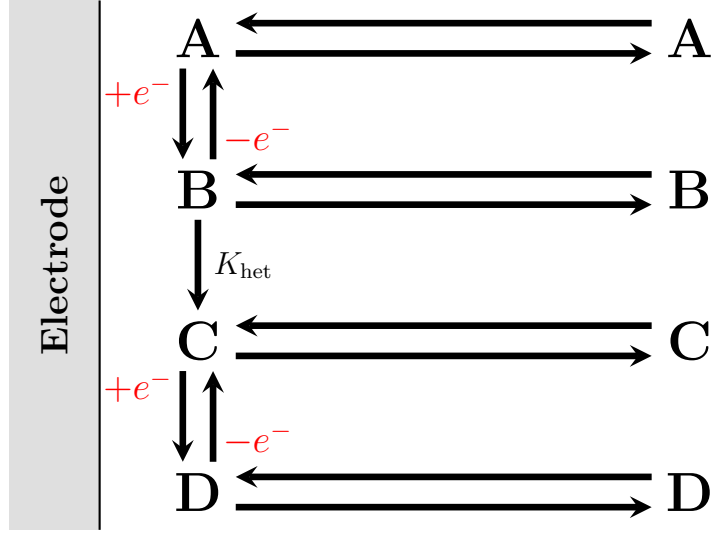


Figure 7.1: Schematic of the $EC_{\text{irr,hetero}}E$ reaction mechanism. Two electrochemical steps are coupled by an irreversible heterogeneous chemical step at the electrode surface.

7.2 Mathematical Description

Under the equal-diffusion assumption ($d_j = 1$), each species satisfies the dimensionless diffusion equation Eq. (2.10). Unlike the single-electron case, the heterogeneous step breaks stoichiometric conservation at the surface since B is consumed both by oxidation to A and by conversion to C. Thus the system cannot be reduced to a single PDE and all four equations must be solved.

7.2.1 Initial and Boundary Conditions

We assume only A is present initially: $C_A(0, X) = 1$, $C_B = C_C = C_D = 0$. These values are also imposed at $X = X_{\text{max}}$.

We apply the scalings of Table 2.1, defining $K_{\text{red}}^{(i)}$ and $K_{\text{ox}}^{(i)}$ for each electron transfer as in Eq. (2.14), and the dimensionless heterogeneous rate constant as $K_{\text{het}} =$

$k_{\text{het}} r_e / D_A$. The boundary conditions at $X = 0$ are [6]

$$\begin{aligned}
\left(\frac{\partial C_A}{\partial X} \right)_{X=0} &= K_{\text{red}}^{(1)} C_A - K_{\text{ox}}^{(1)} C_B, \\
\left(\frac{\partial C_B}{\partial X} \right)_{X=0} &= -K_{\text{red}}^{(1)} C_A + (K_{\text{ox}}^{(1)} + K_{\text{het}}) C_B, \\
\left(\frac{\partial C_C}{\partial X} \right)_{X=0} &= K_{\text{red}}^{(2)} C_C - K_{\text{ox}}^{(2)} C_D - K_{\text{het}} C_B, \\
\left(\frac{\partial C_D}{\partial X} \right)_{X=0} &= -K_{\text{red}}^{(2)} C_C + K_{\text{ox}}^{(2)} C_D,
\end{aligned} \tag{7.3}$$

where all surface concentrations are evaluated at $(T, 0)$. The K_{het} term appears only in the B and C equations and is the sole coupling between the two electron-transfer subsystems.

7.2.2 Current

The measured current includes contributions from both electrochemical steps and the heterogeneous conversion [6]:

$$J = - \left\{ \left(\frac{\partial C_A}{\partial X} \right)_{X=0} + \left(\frac{\partial C_C}{\partial X} \right)_{X=0} + K_{\text{het}} C_B(T, 0) \right\}. \tag{7.4}$$

7.3 Pentadiagonal System

Each species is discretised on the same exponentially expanding mesh of Section 3.1. The four species are coupled only through the boundary conditions at $X = 0$. A species-by-species ordering of the unknowns places the surface coupling entries n positions from the main diagonal.

The bandwidth can be reduced by interleaving the unknowns A with B and C with D . Additionally, A and B are indexed from X_{n-1} down to X_0 , while C and D run from X_0 to X_{n-1} :

$$\mathbf{C} = (C_A^{n-1}, C_B^{n-1}, \dots, C_A^0, C_B^0, C_C^0, C_D^0, \dots, C_C^{n-1}, C_D^{n-1})^T. \tag{7.5}$$

The K_{het} coupling between C_B^0 and C_C^0 is now adjacent and so are all other surface couplings in Eq. (7.3). The result is a pentadiagonal system solved in $O(n)$ per timestep. As noted in Section 5.4, its transpose is again pentadiagonal, so the adjoint system is solved at the same cost.

7.4 Inference Setup

The inference workflow follows Section 6.5. The seven parameters are transformed: $\alpha^{(i)} \mapsto \text{logit}(\alpha^{(i)})$, $K_0^{(i)} \mapsto \log K_0^{(i)}$, $K_{\text{het}} \mapsto \log K_{\text{het}}$, with $\theta_f^{(i)}$ untransformed.

Synthetic data is generated at

$$\phi^* = (0.5, 15.0, 0.2, 0.5, 6.0, 0.4, 25.0)$$

with $\sigma = 1000$, $\eta = 0.02$, $\theta_i = 10.0$ and $\theta_v = 10.0$. Under equal runtime budgets this gives 2,400 NUTS samples and 40,000 RWMH samples.

7.5 Results

7.5.1 Solver Verification

We verify the solver in two limiting cases [26]. When $K_{\text{het}} \rightarrow 0$ the chemical step is inactive and the system reduces to a single electron transfer, whose peak current and potential match Eq. (3.11). When $K_{\text{het}} \rightarrow \infty$ with $K_0^{(2)} \rightarrow \infty$, every molecule of B is instantly converted to C and reduced to D, giving a two-electron transfer with twice the peak current. Figure 7.2 confirms both limits.

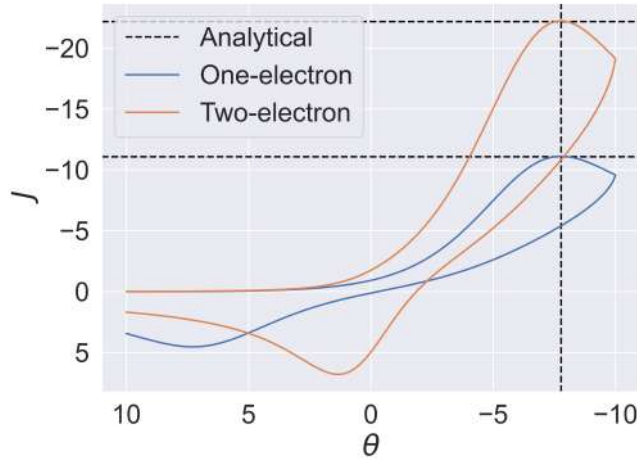


Figure 7.2: Verification in two limiting regimes. Blue: $K_{\text{het}} \rightarrow 0$, recovering the single-electron voltammogram. Orange: $K_{\text{het}} \rightarrow \infty$ with $K_0^{(2)} \rightarrow \infty$, recovering a two-electron transfer. Red dashed lines indicate analytical peak current and potential from Eq. (3.11) (scaled by two for the two-electron case) [26].

7.5.2 Posterior Distributions

Figure 7.3 shows selected pairwise marginal posteriors. Both samplers recover the true values across all seven dimensions; the diagonal overlays both marginals, while the off-diagonal shows the NUTS joint structure alone for clarity. Within the first electron transfer, $\alpha^{(1)}$ and $K_0^{(1)}$ are negatively correlated, which differs from what we saw in the previous chapter. From Eq. (2.14), an increase in $\alpha^{(1)}$ can be compensated by a decrease in $K_0^{(1)}$ to leave $K_{\text{red}}^{(1)}$ near the cathodic peak approximately unchanged; the sign of this trade-off depends on which side of $\theta_f^{(1)}$ the peak sits. The two transfer coefficients $\alpha^{(1)}$ and $\alpha^{(2)}$ are also correlated, propagated through the shared surface coupling between C_B and C_C . Between electron transfers, the rate constants $K_0^{(1)}$ and $K_0^{(2)}$ correlate negatively: an increase in the rate of the first transfer can be offset by a decrease in the second, since both contribute additively to the measured current in Eq. (7.4).

The K_{het} marginal has a width comparable to that of $K_0^{(2)}$, despite its larger value ($K_{\text{het}} = 25$ vs $K_0^{(2)} = 6$). By contrast, $K_0^{(1)}$ has a width of approximately 5 units around its true value of 15. The heterogeneous rate constant thus remains identifiable even at larger values, reflecting the distinct contribution of the chemical step in the current.

Figure 7.4 shows the posterior-mean voltammogram and 95% credible interval against the noisy data; the fit confirms that the sampled posteriors are consistent with the observations.

7.5.3 $\hat{R}$ Convergence and Sampling Efficiency

Figure 7.5 shows $\hat{R}$ over sampling budget. NUTS remains below the 1.01 threshold throughout. RWMH requires a substantially larger fraction of the budget to converge: the $\hat{R}$ curves for $\theta_f^{(1)}$ and $K_0^{(1)}$ remain above 1.01 for the majority of the run.

NUTS achieves 2.1–2.8 $\times$ the ESS of RWMH across all seven parameters (Table 7.1), up from 1.6–1.9 $\times$ in the electron reaction. As the number of parameters grows, a random-walk proposal is increasingly likely to land in low-density regions and must shrink to maintain a reasonable acceptance rate, increasing the within-chain correlation. The gradient-informed proposals do not suffer from this issue.

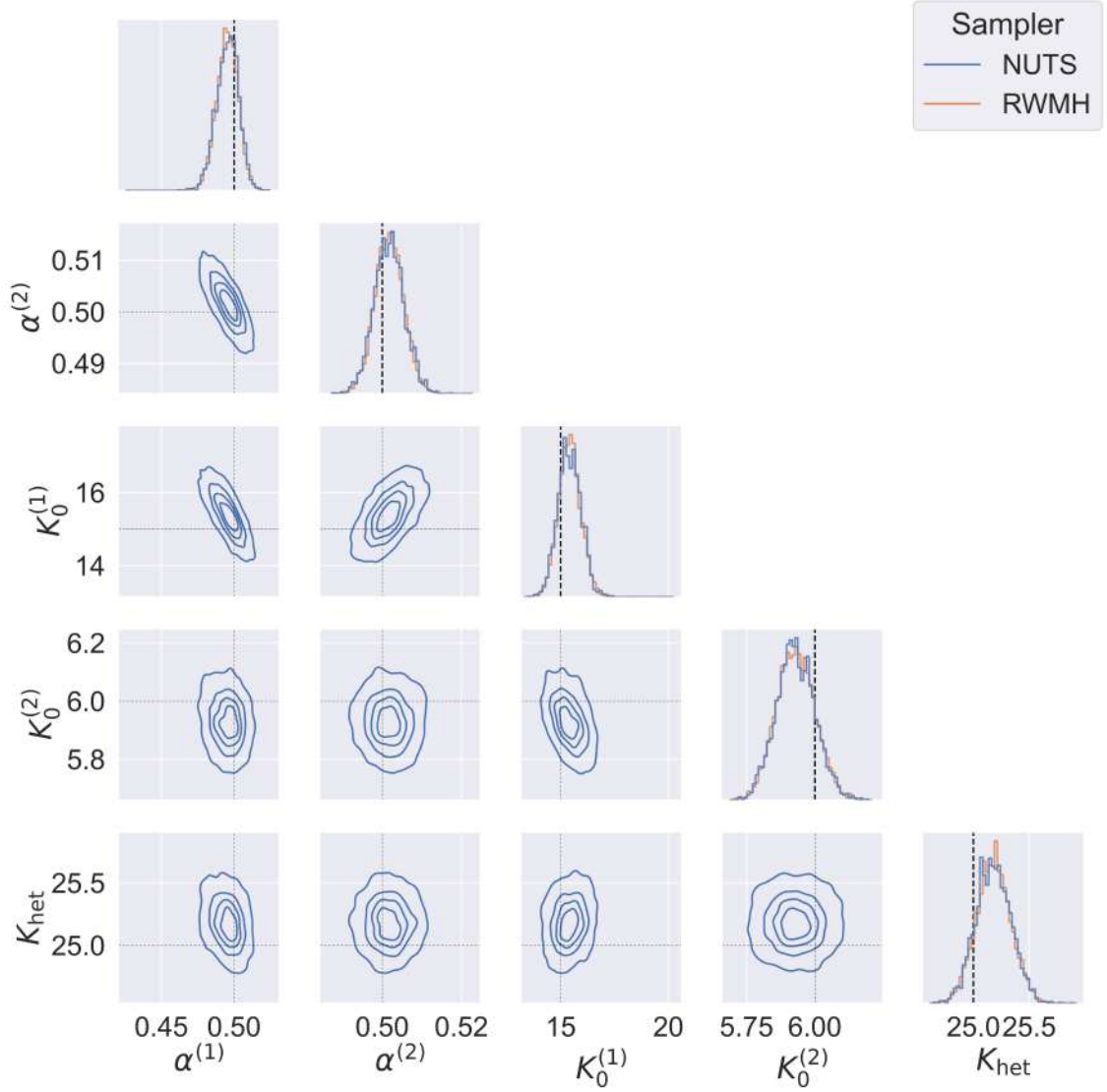


Figure 7.3: Selected pairwise marginal posteriors. Diagonal: histograms of NUTS (blue) with 2,400 samples and RWMH (orange) with 40,000 samples. Off-diagonal: NUTS KDE contours. Dashed lines indicate true values. Formal potentials omitted for clarity.

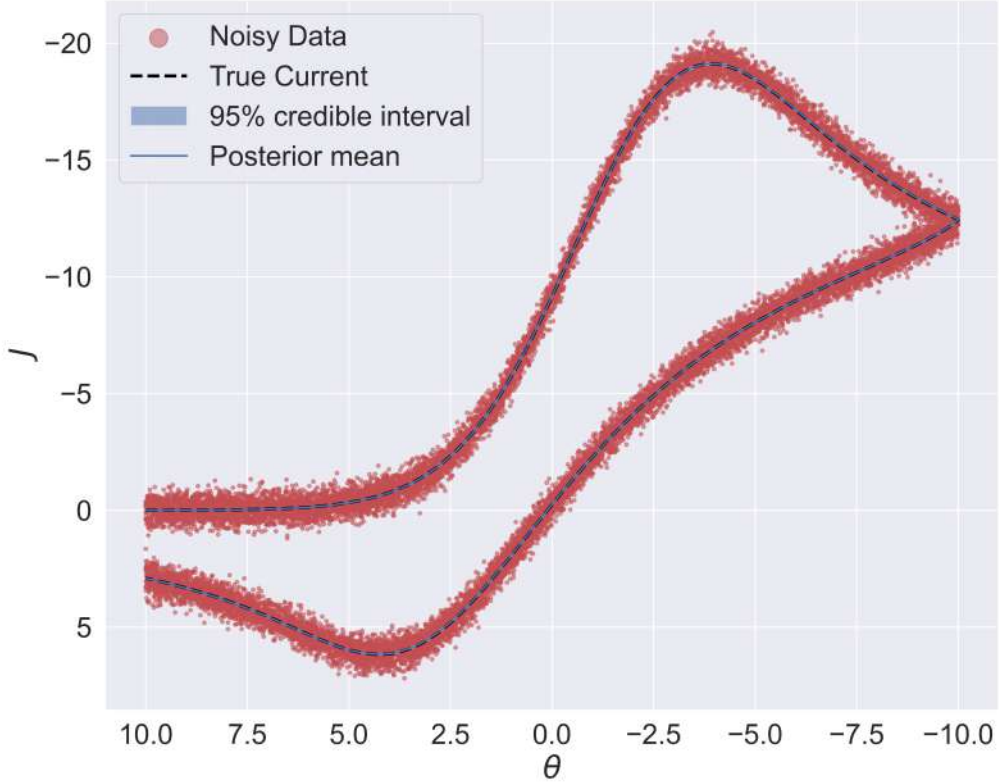


Figure 7.4: Posterior-mean voltammogram (solid) with 95% credible interval (shaded) of the NUTS posterior against the true current (dashed) and noisy data (points). The mean is indistinguishable from the true current with tight credibility intervals.

7.6 AC Voltammetry

In AC voltammetry a sinusoidal perturbation is superimposed on the DC ramp:

$$\theta(T) = \theta_{\text{DC}}(T) + \delta\theta \sin(2\pi f T), \quad (7.6)$$

where $\delta\theta$ is the amplitude and f the frequency. The oscillatory component of the current probes the electrode kinetics on a faster timescale than the DC sweep alone, providing additional information about the rate constants and transfer coefficients. Gavaghan et al. [11] demonstrated this for the single-electron reaction; we extend the comparison to the heterogeneous case.

We use $\delta\theta = 0.5$ and 128 oscillations over the experiment, giving $f = 2 \times 128\pi/T_{\text{max}}$. The 10,000 timesteps from the convergence study resolve the sinusoidal

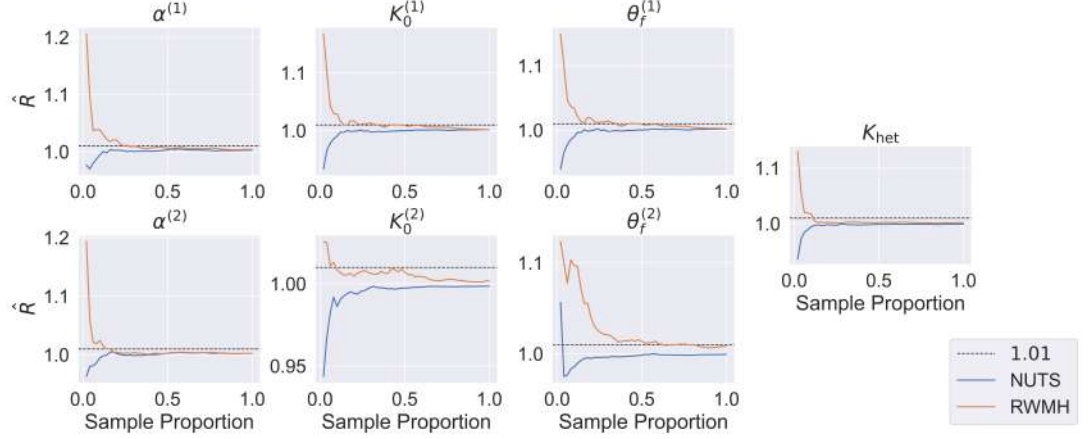


Figure 7.5: $\hat{R}$ as a function of sample proportion for NUTS (blue) and RWMH (orange) for each parameter of the ECirrhetE mechanism. NUTS remains below the 1.01 threshold throughout; RWMH requires a larger fraction of the budget to converge than in the three-parameter case (Fig. 6.5). The dashed line marks the $\hat{R} = 1.01$ convergence threshold [31].

	$\alpha^{(1)}$	$K_0^{(1)}$	$\theta_f^{(1)}$	$\alpha^{(2)}$	$K_0^{(2)}$	$\theta_f^{(2)}$	K_{het}
NUTS	3108.0	2845.3	2841.7	2968.4	3056.3	3869.3	3843.7
RWMH	1347.0	1227.8	1234.4	1408.8	1415.0	1372.3	1404.1
Ratio	2.3	2.3	2.3	2.1	2.2	2.8	2.7

Table 7.1: Effective sample size for each parameter at equal runtime budget, with NUTS/RWMH ratio.

waveform without further refinement. No changes are required to the forward solver or inference workflow.

Figure 7.6 shows the resulting voltammogram at the true parameters; the oscillations are largest near the formal potentials, where the Butler–Volmer kinetics are most sensitive to potential changes.

Figure 7.7 overlays the marginal posteriors from DC and AC data using NUTS. The AC posteriors are narrower for all parameters, with modes more closely aligned with the true values, consistent with Gavaghan et al. [11]. The oscillatory perturbation probes the electrode kinetics at the frequency f , a timescale independent of the DC sweep rate σ , providing additional constraint on the rate constants and transfer coefficients. The DC posteriors are adequate for describing the current, as the results of Fig. 7.4 confirm, but the AC data tighten them further.

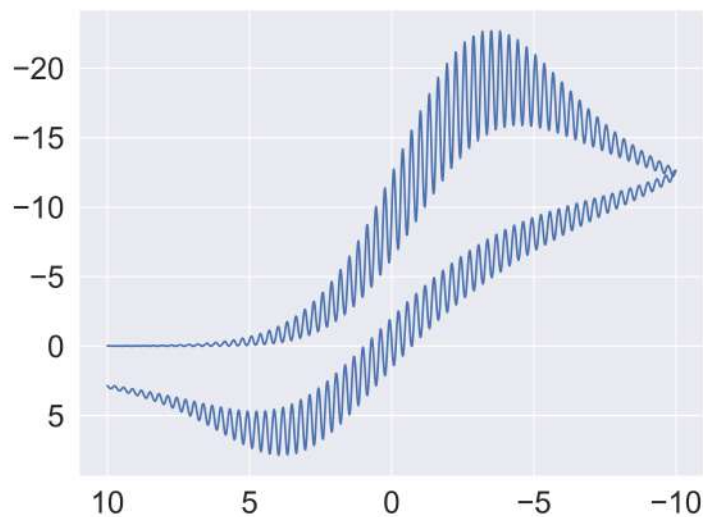


Figure 7.6: AC voltammogram at the true parameter values ($\delta\theta = 0.5$, 128 oscillations). Oscillations are largest near the formal potentials of the two electron transfers.

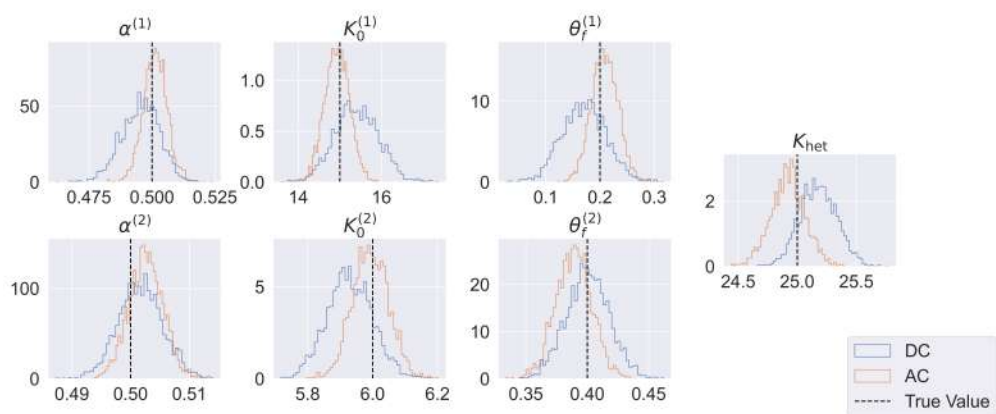


Figure 7.7: Marginal posteriors under DC (blue) and AC (orange) voltammetry. Dashed lines indicate true values. AC posteriors are narrower for all parameters, reflecting additional kinetic information in the oscillatory current.

Chapter 8

Adsorption Reaction

This chapter applies the inference framework to a reaction involving adsorption of the electroactive species onto the electrode surface. Where Chapter 7 treated an irreversible surface step without explicit surface concentrations, the adsorption model introduces surface coverage variables governed by ordinary differential equations coupled to the diffusion PDE through the boundary conditions. Such systems arise experimentally when an electrode is modified with a redox-active film and the voltammetry is performed in a blank electrolyte [6]. The Langmuir adsorption terms contain products of surface coverages and surface concentrations, making the discrete system nonlinear and preventing a direct banded solve. We compare three discretisation strategies – Newton–Raphson, explicit linearisation, and backward implicit linearisation. The parameter space grows to nine dimensions.

8.1 Reaction Scheme

We consider a single electron transfer in which both the reactant A and the product B can adsorb on the electrode surface [6]:



The electron transfer can proceed through both the solution-phase species with rate constants $k_{\text{red/ox}}^{\text{sol}}$. As well as the surface-bound species with rate constants $k_{\text{red/ox}}^{\text{ads}}$. Each species adsorbs onto and desorbs from the electrode surface with rate constants k_{ads}^j and k_{des}^j , competing for a finite number of surface sites set by the maximum surface concentration Γ_{max} (Fig. 8.1).

The solution-phase and adsorbed electron transfers are each characterised by their own transfer coefficient and rate constant: $(\alpha^{\text{sol}}, K_0^{\text{sol}})$ and $(\alpha^{\text{ads}}, K_0^{\text{ads}})$ respectively.

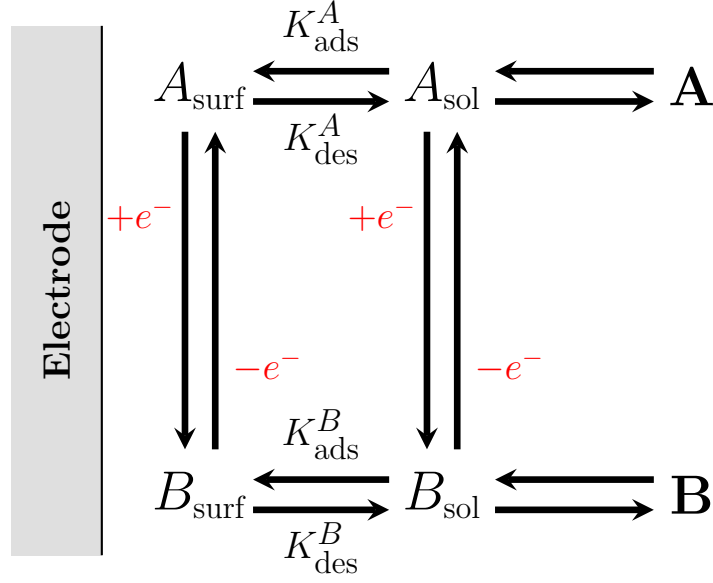


Figure 8.1: Schematic of the adsorption reaction mechanism. Species A and B exist in both a solution phase and a surface-bound phase. Electron transfer can occur between both solution-phase and adsorbed species. Adsorption and desorption exchange species between the two phases.

The nine parameters to be inferred are

$$\phi = (\alpha^{\text{sol}}, K_0^{\text{sol}}, \theta_f^{\text{sol}}, \alpha^{\text{ads}}, K_0^{\text{ads}}, K_{\text{ads}}^{\text{A}}, K_{\text{des}}^{\text{A}}, K_{\text{ads}}^{\text{B}}, K_{\text{des}}^{\text{B}}) \in \mathbb{R}^9. \quad (8.2)$$

8.2 Mathematical Description

8.2.1 Governing Equations

The transport of A and B in solution is governed by the same dimensionless diffusion equation as in Eq. (2.10):

$$\frac{\partial C_j}{\partial T} = \frac{\partial^2 C_j}{\partial X^2}, \quad j \in \{\text{A}, \text{B}\}. \quad (8.3)$$

8.2.2 Surface Coverage Equations

The surface coverages $\xi_j = \Gamma_j/\Gamma_{\text{max}}$ evolve according to

$$\begin{aligned} \frac{d\xi_{\text{A}}}{dT} &= -K_{\text{red}}^{\text{ads}} \xi_{\text{A}} + K_{\text{ox}}^{\text{ads}} \xi_{\text{B}} + K_{\text{ads}}^{\text{A}} C_{0,\text{A}} [1 - (\xi_{\text{A}} + \xi_{\text{B}})] - K_{\text{des}}^{\text{A}} \xi_{\text{A}}, \\ \frac{d\xi_{\text{B}}}{dT} &= K_{\text{red}}^{\text{ads}} \xi_{\text{A}} - K_{\text{ox}}^{\text{ads}} \xi_{\text{B}} + K_{\text{ads}}^{\text{B}} C_{0,\text{B}} [1 - (\xi_{\text{A}} + \xi_{\text{B}})] - K_{\text{des}}^{\text{B}} \xi_{\text{B}}, \end{aligned} \quad (8.4)$$

where $C_{0,j} = C_j(T, 0)$ is the surface concentration in solution. The first two terms in each equation are the adsorbed electron transfer; the third is Langmuir adsorption, with $[1 - (\xi_A + \xi_B)]$ the fraction of unoccupied sites; the fourth is desorption.

8.2.3 Electrode Boundary Conditions

The surface flux of each species is determined by the solution-phase electron transfer and the net adsorption/desorption flux:

$$\begin{aligned} \left(\frac{\partial C_A}{\partial X} \right)_{X=0} &= K_{\text{red}}^{\text{sol}} C_{0,A} - K_{\text{ox}}^{\text{sol}} C_{0,B} + K_{\text{ads}}^A C_{0,A} [1 - (\xi_A + \xi_B)] - K_{\text{des}}^A \xi_A, \\ \left(\frac{\partial C_B}{\partial X} \right)_{X=0} &= \frac{1}{d_B} \left\{ -K_{\text{red}}^{\text{sol}} C_{0,A} + K_{\text{ox}}^{\text{sol}} C_{0,B} + K_{\text{ads}}^B C_{0,B} [1 - (\xi_A + \xi_B)] - K_{\text{des}}^B \xi_B \right\}. \end{aligned} \quad (8.5)$$

The dimensionless rate constants follow the Butler–Volmer form:

$$\begin{aligned} K_{\text{red}}^{\text{sol}} &= K_0^{\text{sol}} \exp[-\alpha^{\text{sol}}(\theta(T) - \theta_f^{\text{sol}})], & K_{\text{ox}}^{\text{sol}} &= K_0^{\text{sol}} \exp[(1 - \alpha^{\text{sol}})(\theta(T) - \theta_f^{\text{sol}})], \\ K_{\text{red}}^{\text{ads}} &= K_0^{\text{ads}} \exp[-\alpha^{\text{ads}}(\theta(T) - \theta_f^{\text{ads}})], & K_{\text{ox}}^{\text{ads}} &= K_0^{\text{ads}} \exp[(1 - \alpha^{\text{ads}})(\theta(T) - \theta_f^{\text{ads}})], \end{aligned} \quad (8.6)$$

where $K_0^{\text{sol}} = k_0^{\text{sol}} r_e / D_A$ and $K_0^{\text{ads}} = k_0^{\text{ads}} r_e^2 / D_A$. The formal potentials of the two couples are related by thermodynamic consistency [6]:

$$\theta_f^{\text{ads}} = \theta_f^{\text{sol}} - \ln \left(\frac{K_{\text{ads}}^A / K_{\text{des}}^A}{K_{\text{ads}}^B / K_{\text{des}}^B} \right). \quad (8.7)$$

θ_f^{ads} is determined by θ_f^{sol} and the four adsorption/desorption dimensionless rate constants already in ϕ which are of the form

$$K_{\text{ads}}^j = \frac{k_{\text{ads}}^j \Gamma_{\text{max}} r_e}{D_A}, \quad K_{\text{des}}^j = \frac{k_{\text{des}}^j \Gamma_{\text{max}} r_e}{c_A^* D_A}. \quad (8.8)$$

8.2.4 Initial Conditions

Initially only species A is present in solution at its bulk concentration: $C_A(0, X) = 1$, $C_B(0, X) = 0$. The initial surface coverages are set by the Langmuir equilibrium:

$$\xi_A(0) = \frac{K_{\text{ads}}^A / K_{\text{des}}^A}{1 + K_{\text{ads}}^A / K_{\text{des}}^A + C_B^* K_{\text{ads}}^B / K_{\text{des}}^B}, \quad \xi_B(0) = \frac{C_B^* K_{\text{ads}}^B / K_{\text{des}}^B}{1 + K_{\text{ads}}^A / K_{\text{des}}^A + C_B^* K_{\text{ads}}^B / K_{\text{des}}^B}. \quad (8.9)$$

We set $C_B^* = 0$ and thus $\xi_B(0) = 0$.

8.2.5 Current

The measured current has contributions from both the solution-phase and adsorbed electron transfers:

$$\frac{I}{\pi r_e F D_A c_A^*} = - \left\{ \left(\frac{\partial C_A}{\partial X} \right)_{X=0} - \frac{1}{\beta} \frac{d\xi_A}{dT} \right\}. \quad (8.10)$$

The first term is the diffusive flux; the second accounts for charge transferred by the conversion of adsorbed A to adsorbed B. The saturation parameter $\beta = c_A^* r_e / \Gamma_{\max}$ is treated as known and fixed at $\beta = 1$.

8.3 Nonlinearity and Discretisation

The Langmuir terms in Eqs. (8.4) and (8.5) contain products of the unknown surface concentrations $C_{0,j}$ and surface coverages ξ_j at the current timestep. After backward Euler discretisation these become products of unknowns at timestep k , making the algebraic system nonlinear in contrast to the first two reactions where all coupling was linear and the system could be solved directly.

8.3.1 Newton–Raphson Solver

We order the unknowns as $\mathbf{x} = (\xi_A^k, \xi_B^k, C_{0,A}^k, C_{0,B}^k, \dots, C_{n-1,A}^k, C_{n-1,B}^k)^T$ and solve the nonlinear residual $\mathbf{F}(\mathbf{x}) = 0$ by Newton–Raphson iteration, using the previous timestep’s solution as the initial guess. After a preliminary row elimination to decouple ξ_A^k and $C_{0,B}^k$, the Jacobian reduces to pentadiagonal form. We solve to tight tolerances ($r_{\text{tol}} = 10^{-10}$, $a_{\text{tol}} = 10^{-12}$) so that this solver serves as the ground-truth reference. Newton–Raphson has a variable number of iterations per timestep for convergence making it incompatible with reverse-mode AD [2]. One could fix the iteration count, but this neither guarantees convergence nor avoids the cost of differentiating through every Newton step. We therefore require a linearised alternative for inference.

8.3.2 Alternative Solvers

Two strategies linearise the nonlinear terms at each timestep, producing systems solvable directly and differentiable through.

Explicit linearisation. The products $C_{0,j}^k [1 - (\xi_A^k + \xi_B^k)]$ are approximated by evaluating the surface coverage at the previous timestep:

$$C_{0,j}^k [1 - (\xi_A^k + \xi_B^k)] \approx C_{0,j}^k [1 - (\xi_A^{k-1} + \xi_B^{k-1})]. \quad (8.11)$$

This is first-order in ΔT and accurate when the surface coverage does not change rapidly between timesteps.

Backward implicit linearisation. Following Britz [3], the product is expanded to first order about the previous-timestep values:

$$C_{0,j}^k \xi_l^k \approx C_{0,j}^k \xi_l^{k-1} + C_{0,j}^{k-1} \xi_l^k - C_{0,j}^{k-1} \xi_l^{k-1}. \quad (8.12)$$

This retains linear coupling to both unknowns at timestep k , also first-order in ΔT but with more implicit coupling than the explicit scheme. Both alternatives produce a pentadiagonal system after row elimination which is compatible with the adjoint framework of Chapter 5.

8.3.3 Solver Comparison

Figure 8.2 compares the two AD-compatible solvers against the Newton reference. The backward implicit scheme achieves errors of order 10^{-7} across most of the sweep, compared with 10^{-3} for the explicit scheme. For both solvers, the error increases when we have a fast change in the gradient of the current.

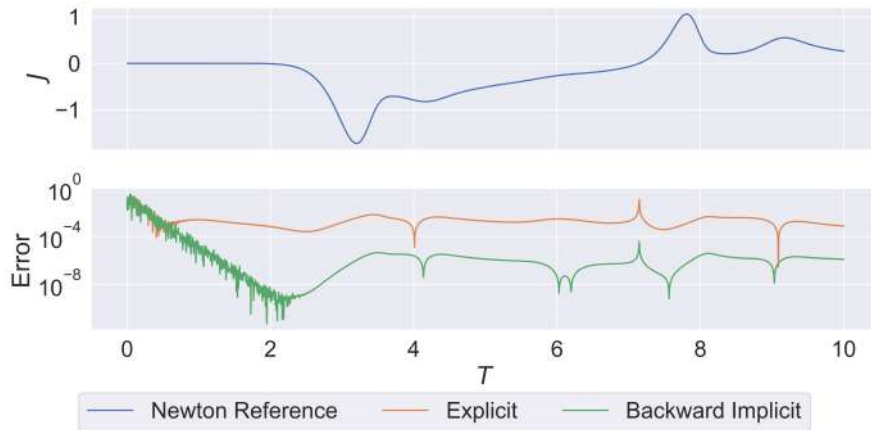


Figure 8.2: Top: Reference Newton–Raphson current against time. Bottom: Absolute relative error of the explicit (orange) and backward implicit (green) solvers against the Newton reference at each timestep. The backward implicit scheme is three to four orders of magnitude more accurate across most of the sweep.

The backward implicit solver is slower than the explicit solver: 80 ms vs 60 ms per forward solve. This 33% overhead is acceptable given the accuracy gain, and we adopt the backward implicit scheme for all inference runs.

8.4 Inference Setup

The inference workflow follows Section 6.5 without modification. The transfer coefficients are logit-transformed, the rate constants log-transformed, and θ_f^{sol} left untransformed. Synthetic data is generated using the Newton solver at the true parameter values

$$\phi^* = (0.5, 10^{-3}, 0.0, 0.45, 0.5, 4.5, 1.0, 1.0, 1.0), \quad (8.13)$$

with dimensionless scan rate $\sigma = 10$, noise level $\eta = 0.02$, and sweep limits $\theta_i = 25.0$ and $\theta_v = -25.0$. The scan rate is two orders of magnitude lower than the preceding reactions since adsorption and desorption operate on a slower timescale than electron transfer, and at high scan rates the potential sweeps too fast for adsorption to respond, rendering the voltammogram insensitive to the adsorption rate constants.

8.5 Results

8.5.1 Solver Verification

We verify the forward solver against the analytical solution for a pure monolayer. This occurs when all electroactive material is pre-adsorbed and no solution-phase electron transfer occurs. For a fully reversible adsorbed redox couple the dimensionless current is [5]

$$J = \mp \frac{\exp(-\theta)}{(1 + \exp(-\theta))^2}, \quad (8.14)$$

where the upper sign refers to the cathodic scan and the lower to the anodic scan. Figure 8.3 confirms that the numerical solution reproduces the full analytical trajectory.

8.5.2 Posterior Distributions

Figure 8.4 shows pairwise marginal posteriors from NUTS (2,400 samples) and RWMH (32,000 samples) under equal computational budgets, with θ_f^{sol} omitted for brevity. Both samplers recover the true parameter values.

α^{sol} and α^{ads} are correlated, similar to the relationship between the transfer coefficients in Section 7.4. The adsorption and desorption rate constants for each species

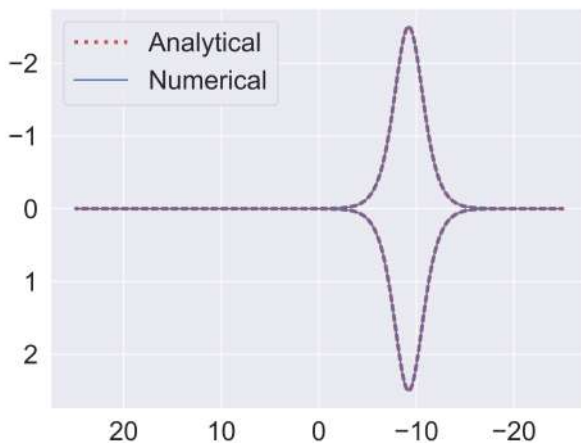


Figure 8.3: Voltammogram for a pure monolayer with a reversible adsorbed redox couple. The numerical solution (solid) matches the analytical current Eq. (8.14) (dashed). $K_0^{\text{sol}} = 0$, $K_0^{\text{ads}} = 10^5$.

are also correlated: the Langmuir terms in Eqs. (8.4) and (8.5) depend on these constants primarily through the equilibrium ratio $K_{\text{ads}}^j/K_{\text{des}}^j$, which sets the surface coverage at steady state. Increasing both constants by the same factor leaves this ratio nearly unchanged, producing a similar voltammogram. The posterior is therefore elongated along the direction in which the ratio is constant. This geometry makes sampling harder: moving along the ridge is easy, but resolving the narrow cross-section requires precise steps, reducing efficiency for both samplers.

Inference uses the backward implicit linearisation, whereas the data were generated with the Newton solver. Both samplers still recover the true parameters, confirming that the linearisation error in Eq. (8.12) is smaller than the observation noise and does not shift the posterior mode.

Figure 8.5 overlays the posterior-mean voltammogram with the true current. The 95% credible interval is tight throughout the sweep but broadens at the various peaks in the voltammogram.

8.5.3 $\hat{R}$ Convergence and Sampling Efficiency

Figure 8.6 shows $\hat{R}$ over sampling budget. NUTS sits below the 1.01 threshold from almost the first checkpoint for all nine parameters. RWMH starts as high as $\hat{R} \approx 1.4$ for K_0^{ads} and requires 20%–30% of the budget to cross the threshold for most parameters.

Table 8.1 gives NUTS/RWMH ESS ratios of 2.6–3.7 $\times$. The electron-transfer parameters (α^{sol} , K_0^{sol} , θ_f^{sol} , α^{ads} , K_0^{ads}) achieve the upper end of this range; the adsorption/desorption rate constants the lower end, reflecting the elongated posteriors along directions of near-constant equilibrium ratio (Fig. 8.4).

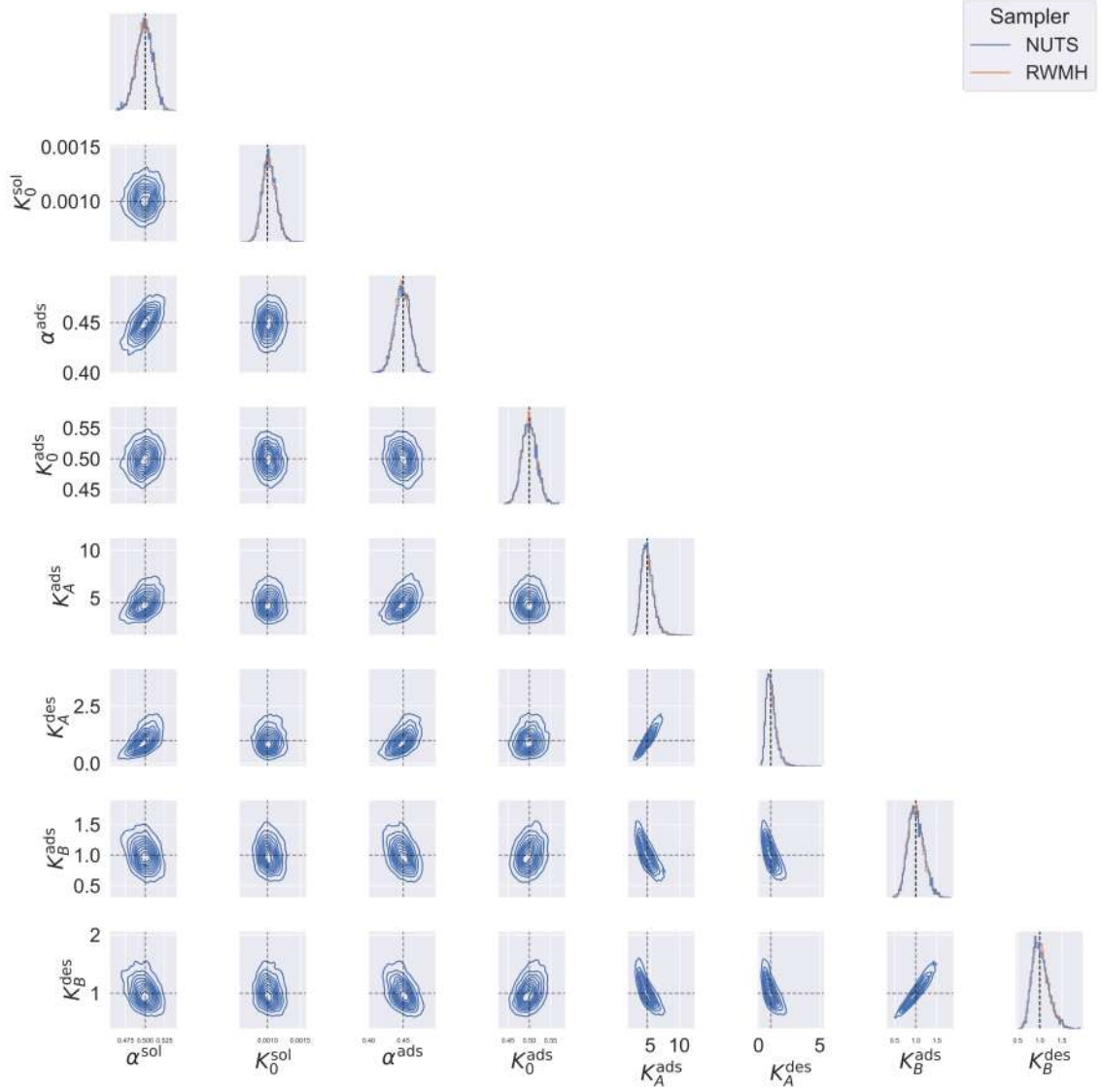


Figure 8.4: Pairwise marginal posteriors for the adsorption reaction. NUTS (blue, 2,400 samples) and RWMH (orange, 32,000 samples) are overlaid; dashed lines indicate the true parameter values. θ_f^{sol} omitted for brevity

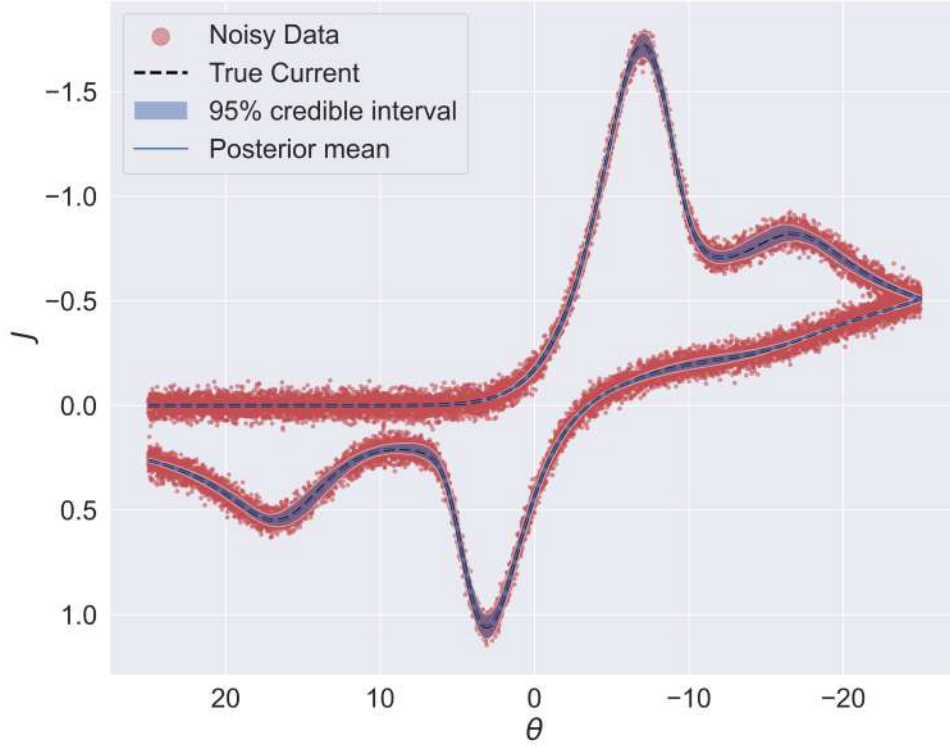


Figure 8.5: Posterior-mean voltammogram (solid) with 95% credible interval (shaded) against the true current (dashed) and noisy data (points).

	α^{sol}	K_0^{sol}	θ_f^{sol}	α^{ads}	K_0^{ads}	$K_{\text{ads}}^{\text{A}}$	$K_{\text{des}}^{\text{A}}$	$K_{\text{ads}}^{\text{B}}$	$K_{\text{des}}^{\text{B}}$
NUTS	3960.4	4470.9	3399.8	3635.7	3728.3	3055.2	2893.7	3558.2	3209.5
RWMH	1152.8	1077.6	1058.6	1252.5	1170.6	1120.7	1063.6	1333.5	1228.0
Ratio	3.4	4.1	3.2	2.9	3.2	2.7	2.7	2.7	2.6

Table 8.1: RWMH and NUTS effective sample size for each parameter at equal sampling runtime, with NUTS/RWMH ratio.

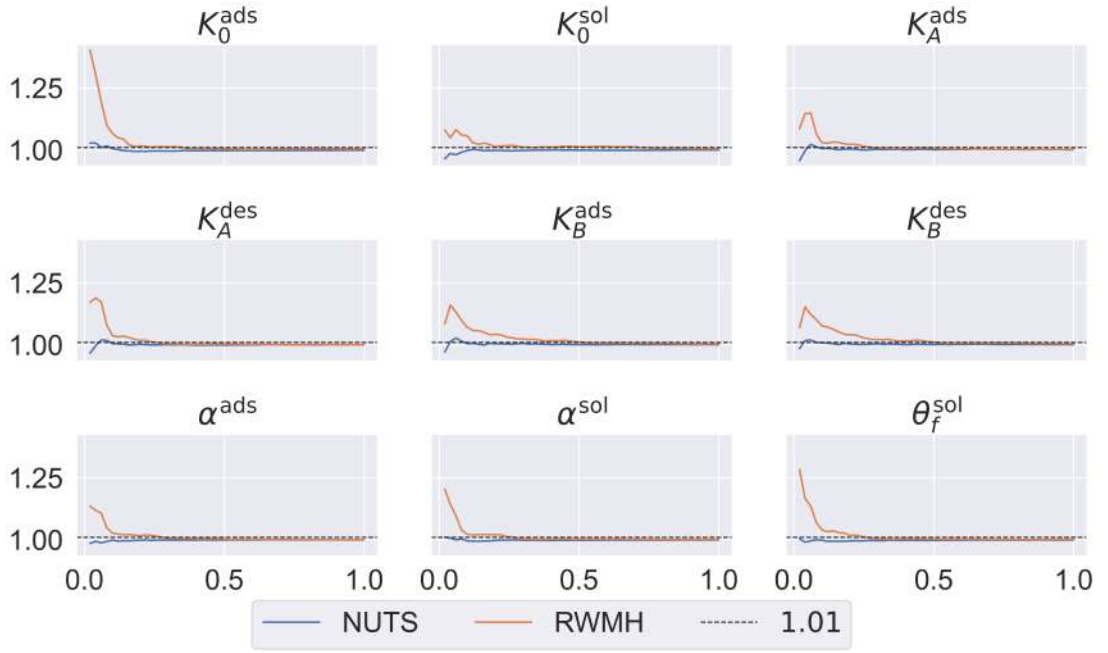


Figure 8.6: $\hat{R}$ as a function of sample proportion for NUTS (blue) and RWMH (orange) for each of the nine parameters of the adsorption mechanism. NUTS remains at the threshold throughout; RWMH requires 20%–30% of the budget to cross $\hat{R} = 1.01$ for most parameters. The dashed line marks the $\hat{R} = 1.01$ convergence threshold [31].

Chapter 9

Conclusion

This dissertation developed a differentiable forward solver for voltammetric simulation and integrated it with gradient-based Bayesian inference to solve the electrochemical inverse problem. The work made three contributions. First, a custom adjoint rule for the banded linear solve yields exact gradients, reducing the per-timestep memory from the full factorisation tape to four arrays of length n (Chapter 5). Second, coupling this solver with NUTS produces full posterior distributions over the physical parameters, quantifying uncertainty and revealing correlations that point estimates cannot provide (Chapters 4 and 6). Third, the framework was applied to three reaction mechanisms of increasing complexity: a single electron transfer, a heterogeneous reaction, and an adsorption reaction (Chapters 6 to 8).

The forward solver for each reaction was verified against analytical solutions before inference. Under matched runtime, NUTS converged in $\hat{R}$ from the first checkpoint for almost every parameter in every reaction, while RWMH required about 30% of the samples before its pooled empirical moments were valid estimates. The NUTS/RWMH ESS ratio grew from 1.6–1.9 $\times$ in three dimensions to 2.1–2.7 $\times$ in seven, and 2.6–4.1 $\times$ in nine, where the lower end of the range corresponds to the adsorption/desorption rate constants with their strongly correlated posteriors.

The adsorption reaction also illustrated how the framework adapts to numerical constraints. The Langmuir coupling terms make the discrete system nonlinear, preventing a direct banded solve. A backward implicit linearisation provided an AD-compatible approximation two to three orders of magnitude more accurate than an explicit alternative, at a 33% runtime overhead. The linearisation error was smaller than the observation noise and did not shift the posterior mode relative to the Newton reference. The adsorption/desorption rate constants exhibited elongated posteriors along directions of constant equilibrium ratio, reducing the NUTS advantage for those parameters specifically.

9.1 Further Work

Experimental data. All results in this dissertation use synthetic data generated from the forward model. Real voltammograms introduce model misspecification: capacitive charging current, ohmic drop through the solution resistance, and electrode surface roughness are absent from the simulations. Applying the framework to experimental data would test the robustness of the posterior distributions under model–data mismatch, and would likely require extending the forward model or introducing nuisance parameters.

Alternative kinetic models. The Butler–Volmer equation was used throughout as the constitutive kinetic law. Marcus–Hush–Chidsey theory [6] provides a more physically grounded model at the cost of a numerical integral in the rate expression. Bayesian model comparison between Butler–Volmer and Marcus–Hush–Chidsey would allow the data to discriminate between kinetic models. Chen et al. [5] have begun this programme with gradient-based optimisation; extending it to full posterior inference is a natural next step.

Neural operator surrogates. The forward solver computes the full concentration field $C(X, T)$, but the likelihood depends only on the surface flux $J(T)$. A neural operator trained on solver output could emulate the parameter-to-current map at a fraction of the cost. Such surrogates are differentiable by construction and NUTS-compatible, and could reduce the per-sample cost by orders of magnitude, making inference over larger parameter spaces or more complex reaction networks practical.

Alternative samplers. NUTS is not the only gradient-based sampler. Micro-canonical Langevin Monte Carlo and other adaptive HMC variants may offer further efficiency gains, particularly for the correlated posteriors encountered in the adsorption case.

The framework developed here is not specific to the three reactions studied. Any voltammetric forward model that can be expressed as a sequence of banded linear solves admits the same adjoint treatment, and the inference workflow applies without modification. As differentiable programming becomes a common methodology in the computational sciences, coupling differentiable PDE solvers with gradient-based samplers offers a systematic route to efficient, full uncertainty quantification for electrochemical inverse problems.

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Roman Ring, Francisco Ruiz, Alvaro Sanchez, Laurent Sartran, Rosalia Schneider, Eren Sezener, Stephen Spencer, Srivatsan Srinivasan, Miloš Stanojević, Wojciech Stokowiec, Luyu Wang, Guangyao Zhou, and Fabio Viola. The DeepMind JAX Ecosystem, 2020.

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Appendix A

Discretisation Convergence

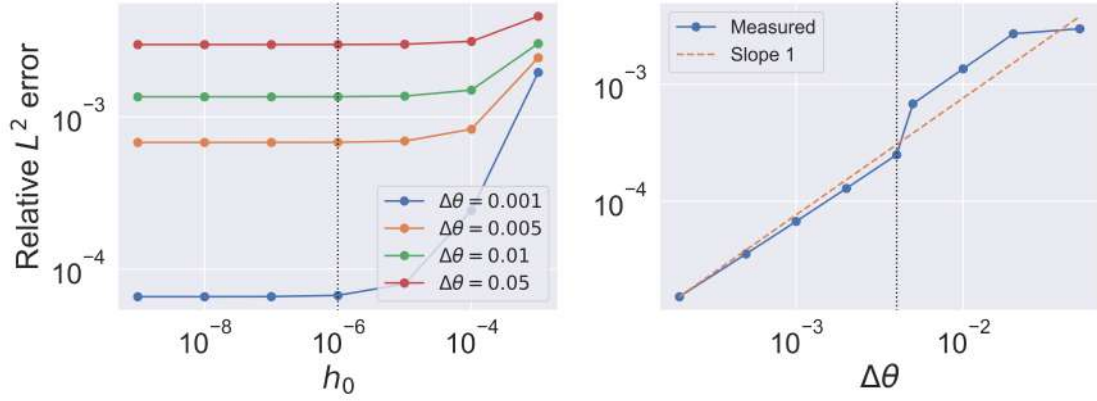


Figure A.1: Convergence of the heterogeneous solver. Relative L^2 error of the simulated current against a reference solution. Left: spatial refinement at four fixed $\Delta\theta$; the error plateaus for $h_0 \lesssim 10^{-6}$. Right: temporal refinement at $h_0 = 10^{-6}$; the measured slope matches the first-order backward Euler rate. Dotted lines mark the adopted values $h_0 = 10^{-6}$ and $\Delta\theta = 4 \times 10^{-3}$.

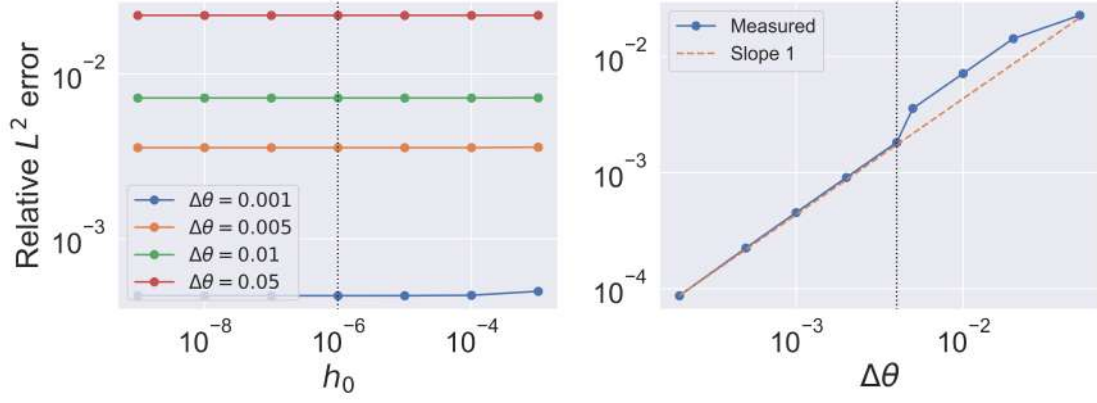


Figure A.2: Convergence of the adsorption solver against a Newton–Raphson reference solution. Left: spatial refinement at four fixed $\Delta\theta$; the error is insensitive to h_0 across the range tested, indicating that the temporal error dominates. Right: temporal refinement at $h_0 = 10^{-6}$; the measured slope matches the first-order backward Euler rate. Dotted lines mark the adopted values $h_0 = 10^{-6}$ and $\Delta\theta = 4 \times 10^{-3}$.

Appendix B

Optimisation Results

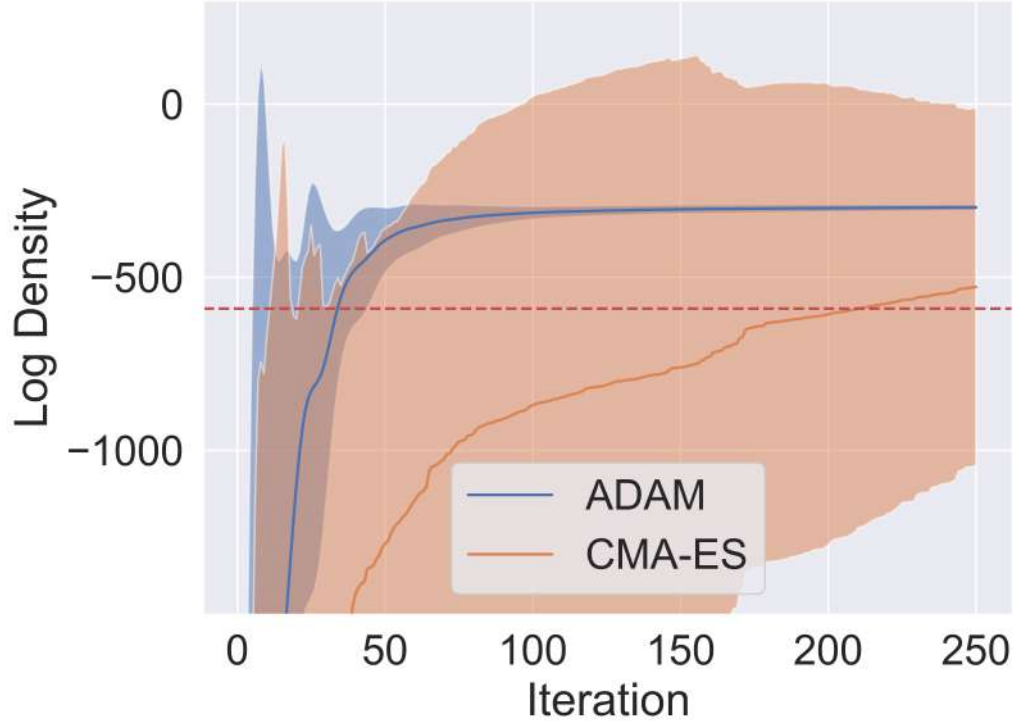


Figure B.1: Log-posterior density during optimisation for ADAM and CMA-ES (population size 9) on the seven-parameter heterogeneous reaction at noise level $\eta = 0.02$. Each algorithm is run from 32 randomly drawn initial conditions; solid lines show the mean and shaded regions indicate ± 1 standard deviation across runs. The black dashed line marks two times the log-density evaluated at the true parameters ϕ^* . Both algorithms are given equal total computational budgets measured in forward-solve equivalents. 225 steps for ADAM and 50 iterations for CMA-ES.

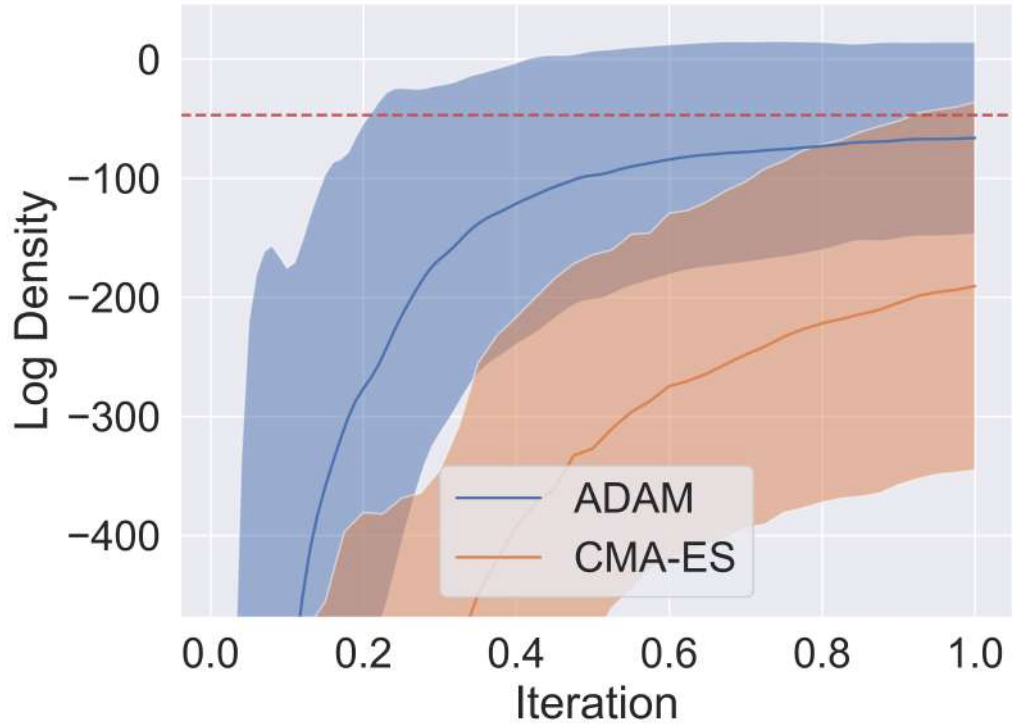


Figure B.2: Log-posterior density during optimisation for ADAM and CMA-ES (population size 10) on the nine-parameter adsorption reaction at noise level $\eta = 0.02$. Each algorithm is run from 32 randomly drawn initial conditions; solid lines show the mean and shaded regions indicate ± 1 standard deviation across runs. The black dashed line marks two times the log-density evaluated at the true parameters ϕ^* . Both algorithms are given equal total computational budgets measured in forward-solve equivalents. 250 steps for ADAM and 50 iterations for CMA-ES.

Appendix C

Custom vector Jacobian Product

```
1 import jax.numpy as jnp
2 from chex import dataclass
3 from jax import custom_vjp, jit
4 from jax.lax import scan
5 from jaxtyping import Scalar
6
7 @dataclass
8 class TriMod:
9     c: Scalar
10    d: Scalar
11
12 @dataclass
13 class TriRow:
14     a: Scalar
15     b: Scalar
16     c: Scalar
17     d: Scalar
18
19 @custom_vjp
20 def tridiagonal_solve(a, b, c, d):
21     return tridiagonal_solve_impl(a, b, c, d)
22
23 def tridiagonal_solve_fwd(a, b, c, d):
24     x = tridiagonal_solve_impl(a, b, c, d)
25     return x, (a, b, c, x)
26
27 def tridiagonal_solve_bwd(res, g):
28     a, b, c, x = res
29     lam = tridiagonal_solve_impl(c, b, a, g)
30     g_b = -lam * x
31     g_a = -lam[1:] * x[:-1]
32     g_c = -lam[:-1] * x[1:]
33     g_d = lam
34     return g_a, g_b, g_c, g_d
35
36 tridiagonal_solve.defvjp(tridiagonal_solve_fwd,
37                          tridiagonal_solve_bwd)
```

```

38 @jit
39 def tridiagonal_solve_impl(
40     a: Scalar,
41     b: Scalar,
42     c: Scalar,
43     d: Scalar,
44 ) -> Scalar:
45     mod_1_denom = b[0]
46     c_mod_1 = c[0] / mod_1_denom
47     d_mod_1 = d[0] / mod_1_denom
48
49     def fwd(mod: TriMod, row: TriRow):
50         mod_denom = row.b - row.a * mod.c
51
52         c_mod = row.c / mod_denom
53         d_mod = (row.d - row.a * mod.d) / mod_denom
54
55         new_mod = TriMod(c=c_mod, d=d_mod)
56         return new_mod, mod
57
58     init_mod = TriMod(c=c_mod_1, d=d_mod_1)
59
60     xs = TriRow(a=a, b=b[1:], c=jnp.concat([c[1:], jnp.zeros(1)]), d
61         =d[1:])
62
63     final_mod, mods = scan(fwd, init_mod, xs)
64
65     def bwd(x_p1: Scalar, mod: TriMod):
66         x = mod.d - mod.c * x_p1
67         return x, x
68
69     x_n = final_mod.d
70     _, x = scan(bwd, x_n, mods, reverse=True)
71
72     solution = jnp.concat([x, jnp.array([x_n])])
73
74     return solution

```

Listing C.1: Implementation of the custom vector Jacobian Product using JAX